

Dynamic generalized network autoregressive modelling of longitudinal networks

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Abstract

Existing models for longitudinal networks are limited: these assume stationary temporal trends and no interactions between groups, impose trivial homogenous group-wise correlation structures, and prohibit analysing the group-wise effects of time-varying covariates. We introduce the dynamic GNAR structural model with interactions for analysing *fixed network effects* that account for temporal and seasonal trends, and an inhomogeneous serial correlation structure. We propose community-driven estimators for small, moderate and large sample settings, model selection criteria, derive finite-sample error bounds for our estimators, and attain asymptotic normality for our fixed network effects estimator. Our results elucidate that empirical process theory can be applied successfully to network time series analysis. We further analyse the dynamics of presidential elections in the USA from 1976 to 2020, that is, a realisation of length twelve on 51 time series (50 states and Washington DC).

Keywords: high-dimensional time series, network time series, presidential elections, finite sample bounds, longitudinal data, network interactions, fixed effects

1 Introduction

Longitudinal data consist of measurements arising from the same unit, usually each unit belongs to a *known group* (e.g., treatment or placebo), collected throughout a time period of length $T > 0$ (Weiss 2005, Chapter 1). If these units are linked to a network, then we say that the data correspond to a longitudinal network. That is, each unit is associated to the same node in a network at all times $t = 1, \dots, T$. In this paper, we introduce the dynamic GNAR structural model, which drastically increases the scope of network time series models for analysing longitudinal network data.

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Our proposed framework surpasses three important limitations: (i) introduces efficient nonparametric estimation of temporal and seasonal trend components, (ii) accounts for serial correlation via a new autoregressive model that allows for inhomogeneous lagged dependence and the inclusion of interaction terms between different groups, and (iii) incorporates a fixed network effects term that quantifies the impact, not necessarily stationary, time-varying covariates might have on each group. These innovations increase the scope of GNAR models and generalise structural models (Shumway 2017, Chapter 6) from the univariate time series setting to the network time series one.

if Y is univariate, then just a MV time series? Why panel or longitudinal?

1.1 Notation and background

Let $Y_{i,t} \in \mathbb{R}$ denote the measurement generated by the i th unit at time $t \in \mathbb{Z}^+$, where there are $i \in [d] := \{1, \dots, d\}$ units. Each one linked to a node in a known or inferred network $\mathcal{G} = (\mathcal{K}, \mathcal{E})$, where $\mathcal{K} = [d]$ is the node set, $\mathcal{E} \subseteq \mathcal{K} \times \mathcal{K}$ is the edge set, and $\mathcal{G}$ is an undirected graph with $d \in \mathbb{Z}^+$ nodes. Then, a first-order *nonstationary* longitudinal network can be generated by the generalised additive model

$$Y_{i,t} = \mu_i(t) + \pi_i(t) + \phi_c v_{i,t} + X_{i,t}, \quad (1)$$

should not \phi depend on i too?

where $\mu_i : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function and $\pi_i : \mathbb{R} \rightarrow \mathbb{R}$ a periodic one that jointly describe node-wise temporal and seasonal trends, $\phi_c \in \mathbb{R}$ is a slope term that incorporates the effects of the time-varying predictors $v_{i,t} \in \mathbb{R}$ for each group $c \in [C]$, where $C \leq d$ is the number of groups, and $X_{i,t}$ is the i th node in a network time series that accounts for serial correlation. We assume that $\phi_c = 0$ if i does not belong to group c . Our objective is to estimate the first-order (trend) and second-order (autocorrelation) structure, and analyse if there are any group-wise differences.

A network time series $\mathcal{X} := (\mathbf{X}_t, \mathcal{G})$ is a stochastic process that manages interactions between nodal time series, i.e., entries in $\mathbf{X}_t \in \mathbb{R}^d$, based on the underlying network. Recently, Knight et al. (2016), Zhu et al. (2017) proposed the generalised network autoregressive (GNAR) model, which provides parsimonious models that often have simpler interpretability *and* superior forecasting performance in a variety of settings, including high-dimensional ones (e.g., Nason & Palasciano 2025). The parsimonious GNAR model allows us to describe a variety of autocorrelation structures, and estimate parameters even if the observations are not equally spaced, or missing, or the realisation length is small compared to the number of unknown parameters, or all of the former. The vector-wise representation of a *global- α* GNAR($p, [s_k]$) model (Nason et al. 2025) is given by

$$\mathbf{X}_t = \sum_{k=1}^p \left(\alpha_k \mathbf{X}_{t-k} + \sum_{r=1}^{s_k} \beta_{kr} \mathbf{Z}_{t-k}^r \right) + \mathbf{u}_t, \quad (2)$$

where $\alpha_k \in \mathbb{R}$ and $\beta_{kr} \in \mathbb{R}$ are the network autoregressive coefficients, $p \in \mathbb{Z}^+$ is the maximum lag, $s_k \in [r^*]$ is the maximum r -stage depth at lag $k = 1, \dots, p$ and $r^* \in [r_{\max}]$ is the maximum r -stage depth across all lags, $r_{\max}$ is the maximum length of finite shortest paths in $\mathcal{G}$. Also, $\mathbf{Z}_t^r := (\mathbf{W}_t \odot \mathbf{S}_r) \mathbf{X}_t$ are r -stage neighbourhood regressions, and $\mathbf{u}_t$ are independent and identically distributed zero-mean white noise errors with covariance matrix $\sigma^2 \mathbf{I}_d$, where $\mathbf{S}_r$ and $\mathbf{W}_t$ are the r -stage adjacency and weights matrix at time t , $\sigma^2 > 0$, and $\odot$ denotes the entry-wise Hadamard product (Nason et al. 2025).

As defined by Knight et al. (2016), nodes i and j are r -stage neighbours if and only if the shortest path between them in $\mathcal{G}$ has a distance of r . The $\mathbf{S}_r$ generalise the notion of adjacency from an edge between nodes to that of shortest paths between them. Define these entry-wise by $[\mathbf{S}_r]_{ij} := \mathbb{I}\{\delta(i, j) = r\}$, where $\delta(i, j)$ is the length of the shortest path between i and j in $\mathcal{G}$, $\mathbb{I}$ is the indicator function, $r \in [r_{\max}]$, and $\delta(i, j) \leq r_{\max}$ for all finite shortest paths. $\mathbf{S}_1$ is the regular adjacency matrix and all the $\mathbf{S}_r$ are symmetric. Let $w_{ij}(t) \in [0, 1]$ be the time-varying edge weights, and denote the $d \times d$ weights matrix by $\mathbf{W}_t$, defined as $[\mathbf{W}_t]_{ij} := w_{ij}(t)$, as proposed by Nason et al. (2025).

GNAR models can perform estimation with *missing data* by setting to zero the weights that involve a missing observation at time t (Knight et al. 2020). If the weights are static, then we drop the t subscript and write $\mathbf{W}$. In the absence of prior weights, GNAR assigns equal importance to each r -stage neighbour at all times t (i.e., $w_{ij} = \{|\mathcal{N}_r(i)|\}^{-1}$, where $|\mathcal{N}_r(i)| \leq d - 1$ is the number of r -stage neighbours of node i and $\mathcal{N}_r(i) \subset \mathcal{K}$ its set of r -stage neighbours). Throughout this work we assume that the network is static. However, GNAR can handle time-varying networks; see Knight et al. (2020). 2016

1.2 Related work

Recent developments for GNAR models include Zhu et al. (2019) for quantiles, Zhou et al. (2020) for Network GARCH models, Armillotta & Fokianos (2024) for Poisson/count data processes, Nason & Wei (2022) to admit time-changing covariate variables, Mantziou et al. (2023) for GNAR processes on the edges of networks and Nason et al. (2025), which introduces the Corbit and R-Corbit plots to aid model selection. Hang Yin & Michailidis (2023) develop a node-wise model and asymptotic theory, but they do not explore community structures and fixed effects, nor do they propose methods for estimating trend and seasonal terms. GNAR models can be seen as constrained VAR ones, which have recently attracted attention for modelling high-dimensional time series; see, e.g., Basu & Michailidis (2015), Basu et al. (2019), Negahban & Wainwright (2011). Nevertheless, these do not exploit nor do they estimate network structures, which have proven useful for increasing model interpretability and outperformed VAR models; see, e.g., Nason & Palasciano (2025). This has made network-based models increasingly popular and led to the development of clustering methods for network linked time series data (Chen et al. 2023, Zhu et al. 2025). However, these methods do not provide tools for de-trending the time series, or consider that the—possibly known—group structure of interest might not be properly captured by distance-based clustering methods for time series data. simple, rather than trivial? or basic?

say multiple stages and lags are often important and mention N+P 2025 paper Both Chen et al. (2023) and Zhu et al. (2025) constrain their models to ~~trivial~~ homogeneous one-lag and one-stage autocorrelation structures with no interactions for all communities. Neither of them define regularised estimators, nor do they provide nonasymptotic error bounds. They further assume that the trend is constant, and derive their asymptotic results by exploiting classic VAR theory. Zhu et al. (2025) restrict their theory to uncorrelated white noise processes. Further, they ignore the case in which group structure is known, or fixed. In contrast, we define a dynamic GNAR structural model that can describe nonstationary longitudinal network data, and propose regularised estimators for small, moderate and large sample sizes.

We emphasize that our work focuses on modelling longitudinal network data with known group structure and temporal trends, *not* on estimating unknown group membership for

a zero-mean stationary network time series. Nevertheless, for completeness, we outline a clustering method based on a new fused group lasso criterion in Appendix E, which extends the methods of Su et al. (2016) and She (2010) for lasso-based (Tibshirani 1996) clustering in regression to network time series.

1.3 Contributions and outline

Our main contribution is introducing the dynamic GNAR structural model that can describe first-order-nonstationary longitudinal network data. Which enables us to estimate nonlinear temporal and seasonal trends, study if time-varying predictors have an effect on network communities, describe inhomogeneous model orders for each community and account for second-order (autocorrelation) interactions between nodes. Section 2 introduces the dynamic GNAR structural and community- α GNAR models with interactions. We propose a new multivariate generalised additive model representation that allows us to derive the *backnet-fitting* algorithm and community-wise penalised least squares estimators for small, moderate and large sample settings. Section 4 presents our main theoretical results. This work is, to the best of our knowledge, the first to derive finite-sample tail bounds on estimators for these multivariate processes. Section 6.2 presents a new longitudinal analysis for Bruma.

Bruma?

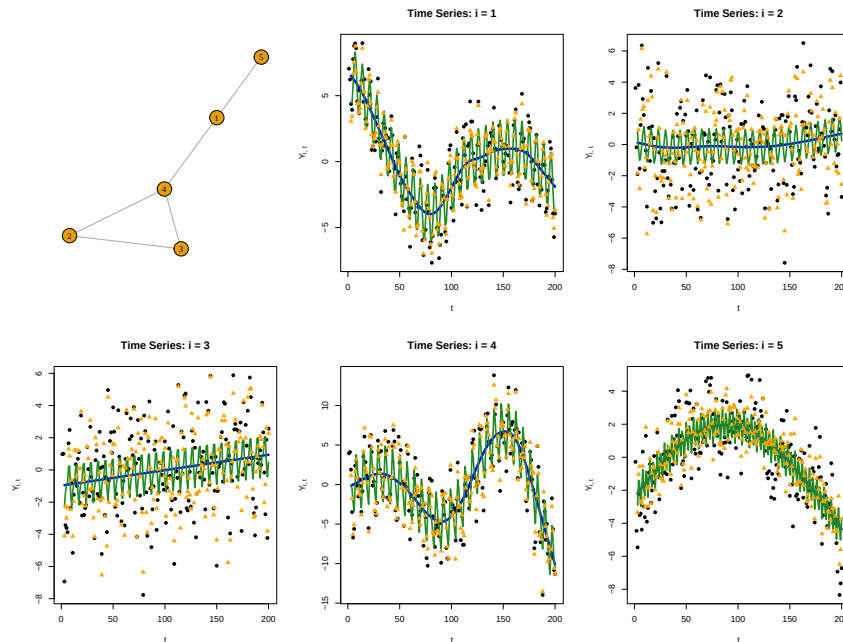


Figure 1: Illustration of a dynamic GNAR structural model. The first plot is the **fiveNet** network included in the **GNAR** package (Knight et al. 2024). The latter are nodal time series realisations, each of length 200 generated by (1). The blue curve is temporal trend, green is seasonal, and the orange points are trend plus fixed effects (fitted points) .

2 The parsimonious GNAR family

The GNAR modelling framework produces parsimonious models that can be thought of as network-informed reparameterizations of VAR models; see Nason et al. (2025). Crucially, the number of parameters in GNAR models grows linearly with dimension, whereas it grows in polynomial fashion for VAR models; see Nason & Wei (2022). Further, the GNAR framework is network-agnostic, that is, we do not impose any constraints (e.g., stochastic block model assumptions Zhu et al. 2025) on the process that generates the network.

2.1 Dynamic GNAR structural models

The dynamic GNAR structural model is given by the generalised additive model

$$Y_{i,t} = \mu_i(t) + \pi_i(t) + \phi_c^T \mathbf{v}_{i,t} + X_{i,t}, \quad (3)$$

where $\mathbf{Y}_t = (Y_{1,t}, \dots, Y_{d,t})$ is a d -dimensional response variable, $\mathbf{v}_{i,t} \in \mathbb{R}^{q_v}$ is a q_v -dimensional, not necessarily stationary, predictor with unknown coefficients $\phi_c = \{\phi_{c,1}, \dots, \phi_{c,q_v}\}$ for each community $c \in [C]$, and $\mu_i(t) \in \mathbb{R}$ is the temporal trend and $\pi_i(t) \in \mathbb{R}$ the seasonal one of unit i . Above in (3), $\mathbf{X}_t$ is a GNAR process linked identically to the same network, i.e., node i is linked to the same series for $\mathbf{Y}_t$ and $\mathbf{X}_t$. For identifiability purposes, we adopt the convention of no intercept (i.e., none of the terms in (3) has an intercept), and $\phi_c = 0$ if $i \notin K_c$. The term $\phi_c^T \mathbf{v}_{i,t}$ in (3) measures the community-wise *fixed network effects* $\mathbf{v}_t$ has on K_c . The trend and seasonal components are unknown *functions*, rather than unknown constants or identically zero.

We assume that $\mathbf{X}_t$ is a zero-mean GNAR process and ^{can} write (3) as the multivariate generalised additive model process,

$$\mathbf{Y}_t = \boldsymbol{\mu}(t) + \boldsymbol{\pi}(t) + \mathbf{V}_t \boldsymbol{\phi}_{[C]} + \mathbf{R}_t \boldsymbol{\theta} + \mathbf{u}_t, \quad (4)$$

where $\boldsymbol{\mu}(t) = \{\mu_1(t), \dots, \mu_d(t)\}$, and $\boldsymbol{\pi}(t) = \{\pi_1(t), \dots, \pi_d(t)\}$ are nonlinear trends, $\mathbf{u}_t$ are zero-mean white noise, such that the noise vectors $\mathbf{u}_t, \mathbf{u}_s$ are uncorrelated at all different times, i.e., $\text{cov}(\mathbf{u}_t, \mathbf{u}_s) = \mathbb{I}(t = s) \odot \boldsymbol{\Sigma}_u$. The “design” matrix $\mathbf{R}_t$ is defined by the linear model representation of a GNAR process, which is given below by (11). The time-varying, predictor design is the $d \times Cq_v$ block-diagonal matrix $\mathbf{V}_t$, such that for each $c \in [C]$, we have $[\mathbf{V}_t \boldsymbol{\phi}_{[C]}]_i = \phi_c^T \mathbf{v}_{i,t}$ for all $i \in K_c$, where $\boldsymbol{\phi}_{[C]} := (\phi_1, \dots, \phi_C)$.

2.2 The community- α GNAR model

Suppose that there is a collection of covariates $c \in [C]$ such that the i th time series $X_{i,t}$ is linked to only one covariate at all times $t \in \mathbb{Z}_0^+$, where $C \leq d$ is the number of covariates. Let $K_c := \{i \in \mathcal{K} : X_{i,t} \text{ is associated with covariate } c\}$ be disjoint node communities defined by the *known* covariates. Intuitively, each covariate is a label that indicates the cluster to which $X_{i,t}$ belongs. For example, if $\mathcal{G}$ consists of population centres, then each $X_{i,t}$ could be characterised as either urban or rural, or a hub-town.

We select the c -community defined by K_c using the vectors $\mathbf{a}_c \in \mathbb{R}^d$. These are given by $\mathbf{a}_c := (a_{1,c}, \dots, a_{d,c})$, where $a_{i,c} := \mathbb{I}(i \in K_c)$, i.e., each entry in $\mathbf{a}_c$ is non-zero if and only if $i \in K_c$. The c -community r -stage neighbourhood regressions at lag k are

$$\mathbf{Z}_{t-k}^{r,c} := (\mathbf{W}_{t,c} \odot \mathbf{S}_r) \mathbf{X}_{t-k}^c, \quad \text{where } \mathbf{X}_t^c := \mathbf{a}_c \odot \mathbf{X}_t, \quad (5)$$

and $\mathbf{W}_{t,c}$ is the c -community weights matrix, which is $\mathbf{W}_t$ constrained to weights between nodes in the same community, i.e., $[\mathbf{W}_{t,c}]_{ij} = w_{ij}(t) \mathbb{I}(i \in K_c \text{ and } j \in K_c)$, and $\mathbf{S}_r$ is the r -stage adjacency matrix.

Let $\mathcal{I}_c := \{\tilde{c} \in [C] : c \text{ interacts with } \tilde{c}, \text{ where } \tilde{c} \neq c\}$. Then, $\mathcal{I}_c$ is the set that indicates which communities have an effect on K_c (i.e., the interactions set for K_c). Note that $\tilde{c} \in \mathcal{I}_c$ does not necessarily require that $c \in \mathcal{I}_{\tilde{c}}$, $\mathcal{I}_c \subset [C]$ and $|\mathcal{I}_c| \leq C - 1$ for all $c \in [C]$. The effect community $K_{\tilde{c}}$ has on community K_c , at the k th lag and r th r -stage, is modelled by the $(c : \tilde{c})$ -community r -stage neighbourhood regressions,

$$\mathbf{Z}_{t-k}^{r,c;\tilde{c}} = a_c \odot (\mathbf{W}_t \odot \mathbf{S}_r) \mathbf{X}_{t-k}^{\tilde{c}}, \quad (6)$$

where each entry $Z_{i,t-k}^{r,c;\tilde{c}}$ is equal to the r -stage neighbourhood regression for node i restricted to nodes that belong to community $K_{\tilde{c}}$.

Let $p_c \in \mathbb{Z}^+$ and $s_k(c) \in \{1, \dots, r_c^*\}$, where the former denotes maximum lag and the latter maximum r -stage depth at lag $k = 1, \dots, p_c$ for community K_c . Note that r_c^* is the maximum r -stage depth for community K_c , and as above, C is the number of non-overlapping communities that are characterised by the covariates $c \in [C]$.

The community- α GNAR model with interactions is given by the sum of

$$\alpha_c(\mathbf{X}_t) := \sum_{k=1}^{p_c} \alpha_{k,c} \mathbf{X}_{t-k}^c, \quad \beta_c(\mathbf{X}_t) := \sum_{k=1}^{p_c} \sum_{r=1}^{s_k(c)} \beta_{k,r,c} \mathbf{Z}_{t-k}^{r,c}, \quad \gamma_c(\mathbf{X}_t) := \sum_{k=1}^{p_c} \sum_{r=1}^{s_k(c)} \sum_{\tilde{c} \in \mathcal{I}_c} \gamma_{k,r,c;\tilde{c}} \mathbf{Z}_{t-k}^{r,c;\tilde{c}}, \quad (7)$$

where $\alpha_{k,c}$, $\beta_{k,r,c}$ are the c -community autoregressive coefficients and $\gamma_{k,r,c;\tilde{c}} \in \mathbb{R}$ is the interaction coefficient at lag k and r -stage equal to r , and $\mathbf{Z}_{t-k}^{r,c}$ and $\mathbf{Z}_{t-k}^{r,c;\tilde{c}}$ are neighbourhood regressions. The former is given by (5) and the latter by (6). We express the community- α GNAR model with interactions in compact structural form as

$$\mathbf{X}_t = \sum_{c=1}^C \{\alpha_c(\mathbf{X}_t) + \beta_c(\mathbf{X}_t) + \gamma_c(\mathbf{X}_t)\} + \mathbf{u}_t, \quad (8)$$

where $\mathbf{u}_t$ are zero-mean independent and identically distributed white noise such that $\text{cov}(\mathbf{u}_t) = \sigma^2 \mathbf{I}_d$ and $\sigma^2 > 0$.

We denote model order as community- α GNAR($[p_c]$, $\{[s_k(c)]\}$, $[C]$, $\mathcal{I}[C]$), where p_c , $s_k(c)$ specify model order for each community, C is the number of community-identifying covariates, and $\mathcal{I}_c$ is the set of communities that interact with K_c . If there are no interaction terms, i.e., $\mathcal{I}[C] = \emptyset$, then $\gamma_c(\mathbf{X}_t) \equiv 0$. The model in (8) can be extended by specifying different model orders for the interaction terms, and including predictor variables to (7) is straightforward. We leave these extensions for future work.

2.2.1 Conditions for stationarity

Remark 1 shows that (8) can be expressed as a VAR by imposing networked-informed constraints on the autoregressive matrices.

Remark 1. Let $\Phi_k(t)$ be the $d \times d$ matrix given by

$$\Phi_k(t) = \sum_{c=1}^C \left[\text{diag}(\alpha_{k,c} \mathbf{a}_c) + \sum_{r=1}^{s_k(c)} \left\{ \beta_{k,r,c} (\mathbf{W}_{t,c} \odot \mathbf{S}_r) + \sum_{\tilde{c} \in \mathcal{I}_c} \gamma_{k,r,c;\tilde{c}} (\mathbf{W}_{t,c;\tilde{c}} \odot \mathbf{S}_r) \right\} \right], \quad (9)$$

where terms for larger model order are set to zero, e.g., if $p_c < p_{\bar{c}}$, then $\alpha_{k,c} \equiv 0$ for $k > p_c$, $[\mathbf{W}_{t,c}]_{ij} = w_{ij}(t)\mathbb{I}(i \text{ and } j \in K_c)$, and $[\mathbf{W}_{t,c;\bar{c}}]_{ij} = w_{ij}(t)\mathbb{I}(i \in K_c \text{ and } j \in K_{\bar{c}})$. Then, the VAR(p) model given by

$$\mathbf{X}_t = \sum_{k=1}^p \Phi_k(t-k) \mathbf{X}_{t-k} + \mathbf{u}_t, \quad (10)$$

where $\mathbf{u}_t$ are i.i.d. white noise, is identical to the GNAR model given by (8).

In view of Remark 1, stationarity conditions follow immediately from the ones for VAR processes (Lütkepohl 2005, Chapter 2). Interestingly, community- α GNAR models can be thought of as being “between” a local- $\alpha\beta$ and a global- α GNAR specification *and* incorporate community structure into GNAR correlation structures. For all models introduced in Section 2, we outline conditions for stationarity in Appendix C.

3 Model estimation for GNAR models

Let $\boldsymbol{\theta}_c := (\alpha_{k,c}, \beta_{k,c}, \gamma_{k,c}) \in \mathbb{R}^{q_c}$, for $k = 1, \dots, p_c$, where $\beta_{k,c} = (\beta_{k,r,c})$, and for each $r = 1, \dots, s_k(c)$, the vector $\gamma_{k,c}$ concatenates $\gamma_{k,r,c;\bar{c}}$ in ascending order with respect to $\bar{c} \in [C]$. We define $\boldsymbol{\theta} := (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_C) \in \mathbb{R}^q$, where $q = \sum_{c=1}^C q_c$ is the number of unknown parameters. Then, each time step can be expressed as a d -dimensional linear model with “design” matrix $\mathbf{R}_t \in \mathbb{R}^{d \times q}$, where each column is equal to either an autoregressive, between community or within community term in (8). That is, the network-informed design process is

$$\mathbf{R}_t \boldsymbol{\theta} = \sum_{c=1}^C \{\alpha_c(\mathbf{X}_t) + \beta_c(\mathbf{X}_t) + \gamma_c(\mathbf{X}_t)\}, \quad (11)$$

for all $t \geq p + 1$. Thus, we can write (8) as a linear model,

$$\mathbf{X}_t = \mathbf{R}_t \boldsymbol{\theta} + \mathbf{u}_t, \quad (12)$$

where $\mathbf{u}_t$ are i.i.d. for all $t = p + 1, \dots, T$ as in (8).

Assume that we observe $T \in \mathbb{Z}^+$ time-steps of a stationary community- α GNAR process with known order as defined in Section 2. Note that we can write $\mathbf{R}_t$ as row-wise concatenated community-wise “design” matrices, i.e., $\mathbf{R}_t = [\mathbf{R}_{t,1}, \dots, \mathbf{R}_{t,C}]$, and that there are $n_c := T - p_c$ “response” values $\mathbf{y}_{t_i,c} := \mathbf{a}_c \odot \mathbf{X}_t$ for each community, where $t_i \in [t_{n_c}]$ and $t = p + 1, \dots, T$ are the “active” time steps. Then, we concatenate the $\mathbf{R}_{t_i}$ and $\mathbf{y} := (\mathbf{y}_{t_1}, \dots, \mathbf{y}_{t_n}) \in \mathbb{R}^{nd}$, i.e., the ‘response vector’ for the whole model, to obtain

$$\mathbf{y} = \mathbf{R} \boldsymbol{\theta} + \mathbf{u}, \quad (13)$$

where $\mathbf{u}$ is a vector of independent and identically distributed zero-mean random variables.

Following the above, we estimate $\boldsymbol{\theta}$, unless stated otherwise, by least squares,

$$\hat{\boldsymbol{\theta}} = (\mathbf{R}^T \mathbf{R})^{-1} \mathbf{R}^T \mathbf{y}. \quad (14)$$

Remark 2. $\mathbf{R}^T \mathbf{R}$ is community-wise orthogonal (each block corresponds to one community), and since $\text{cov}(\hat{\boldsymbol{\theta}}) = (\mathbf{R}^T \mathbf{R})^{-1}$, the $\hat{\boldsymbol{\theta}}_c$ are uncorrelated and non-zero entries in the precision matrix, $\{\text{cov}(\hat{\boldsymbol{\theta}})\}^{-1}$, correspond to estimated coefficients in the same community.

The estimator given by (14) is an M-estimator; see Proposition 3 in Section K.3 of the supplementary material. By Remark 2, it is reasonable to estimate model parameters separately. This allows us to use more observations for communities with a smaller maximum lag, remove unnecessary predictors, and perform estimation in parallel. In practice, we are often interested in data with correlated errors, so we propose an alternating algorithm for simultaneous covariance and parameter estimation.

Suppose that $\text{cov}(\mathbf{u}_t) = \boldsymbol{\Sigma}_{\mathbf{u}}$ for $\mathbf{u}_t$ in (8) at all times t . If we knew $\boldsymbol{\Sigma}_{\mathbf{u}}$, then we could estimate $\boldsymbol{\theta}$ by generalised least squares,

$$\hat{\boldsymbol{\theta}}_{\text{gls}} = (\mathbf{R}^T \boldsymbol{\Sigma}_T^{-1} \mathbf{R})^{-1} \mathbf{R}^T \boldsymbol{\Sigma}_T^{-1} \mathbf{y}, \quad (15)$$

where $\boldsymbol{\Sigma}_T = \mathbf{I}_n \otimes \boldsymbol{\Sigma}_{\mathbf{u}}$, and each block corresponds to one community. Algorithm 1 in Appendix D estimates $\boldsymbol{\Sigma}_{\mathbf{u}}$ and $\boldsymbol{\theta}$ simultaneously via an alternating procedure; see Chapter 17 in Hastie et al. (2017) for an overview of alternating algorithms.

Moreover, the linear model in (13) can be broken into its community components, which can be estimated by different strategies. We select a loss function, say $\mathcal{L}_c$, for each community, and solve accordingly for $\boldsymbol{\theta}_c$,

$$\hat{\boldsymbol{\theta}}_c = \underset{\boldsymbol{\theta}_c}{\text{argmin}} \{ \mathcal{L}_c(\mathbf{y}_c, \mathbf{R}_c; \boldsymbol{\lambda}) \}, \quad (16)$$

where $\boldsymbol{\lambda}$ is a vector of hyperparameters. In this work we propose two extensions, one uses community-wise elastic-net (Zou & Hastie 2005) estimation,

$$\mathcal{L}_c(\mathbf{y}_c, \mathbf{R}_c; \boldsymbol{\lambda}) = \|\mathbf{y}_c - \mathbf{R}_c \boldsymbol{\theta}_c\|_2^2 + \lambda_{\text{lasso}} \|\boldsymbol{\theta}_c\|_1 + \lambda_{\text{ridge}} \|\boldsymbol{\theta}_c\|_2^2, \quad (17)$$

where $\boldsymbol{\lambda} = (\lambda_{\text{lasso}}, \lambda_{\text{ridge}}) \succeq 0$. An alternating algorithm that minimises (17) is provided in Appendix F of the supplementary material. Many time series data contain large deviations from the mean likely to be outliers, so we also propose a robust estimator based on a community-wise Huber (Huber & Ronchetti 2011) loss; see Appendix F.

Many popular loss functions are a penalised form of least squares, which facilitates extensions that exploit (12), e.g., (17). We highlight that as long as the penalty terms are separable by community (i.e., the penalty term only depends on parameters for said community, e.g., $\|\boldsymbol{\theta}_c\|_p$), then it is valid to estimate $\boldsymbol{\theta}$ either in parallel or an alternating manner; see Chapter 5 in Hastie et al. (2015).

3.1 The high-dimensional case

Many longitudinal network data can be characterised as high-dimensional, i.e., the effective sample size is smaller than the number of unknown parameters, in our notation $nd < q$. The parsimonious nature of GNAR models mitigates this. However, for multivariate time series linked to a large network ($d \gg T$), it might be necessary to use regularised estimators.

Let $\mathcal{L}_{\Sigma}$ be a loss function linked to a penalised covariance estimator, e.g., the graphical lasso (Friedman et al. 2008), usually the input to this procedure will be the residuals from some GNAR fit (i.e., $\hat{\mathbf{u}}_{t_i} = \mathbf{y}_{t_i} - \mathbf{R}_{t_i} \hat{\boldsymbol{\theta}}$). Further, denote by $\mathcal{L}_{\boldsymbol{\theta}}$ a penalised least squares loss function for estimating $\boldsymbol{\theta}$, e.g., the elastic-net in (17). Algorithm 5 exploits this connection and generalised least squares (15) to simultaneously estimate $\boldsymbol{\Sigma}_{\mathbf{u}}$ and $\boldsymbol{\theta}$ when $nd < q$.

We highlight that GNAR models are not limited to these choices of covariance estimators, it is possible to incorporate structured covariance forms into (12), such as parameterized covariances $\Sigma_u(\rho)$, where ρ parameterizes Σ_u (Weiss 2005, Chapter 8). Then, this covariance structure and its estimation procedure can be used in the covariance estimation step in either Algorithm 1 or 5.

3.2 Model estimation for dynamic GNAR models

Ok, so this is new...

Assume that we observe a realisation of length T generated by (4), and let $n = T - p$, where p is lag-order for the GNAR term. Then, analogously to Section 3, we can concatenate the realisation into a generalised additive model,

$$\mathbf{y} = \boldsymbol{\mu}_{(t_1:t_n)} + \boldsymbol{\pi}_{(t_1:t_n)} + \mathbf{V}\boldsymbol{\phi}_{[C]} + \mathbf{R}\boldsymbol{\theta}_X + \mathbf{u}, \quad (18)$$

where $\boldsymbol{\mu}_{(t_1:t_n)} = \text{vec}\{\mu_1(t_i), \dots, \mu_1(t_i)\}$ for $t_i = t_1, \dots, t_n$ denotes temporal trend values, and $\boldsymbol{\pi}_{(t_1:t_n)}$ seasonal ones, and $\mathbf{R}$ is the GNAR design matrix from (13).

We highlight that the $nd \times Cq_v$ matrix $\mathbf{V}$ is the predictor design matrix, and $\boldsymbol{\phi}_{[C]}$ is the Cq_v -dimensional vector of fixed network effects. Further, $\boldsymbol{\theta}_X$ is the q -dimensional vector of unknown autoregressive coefficients of the GNAR process, as defined in Section 3, and $\mathbf{u}$ is the same error vector as in (12). Motivated by generalised additive models (Wood 2017), we propose a new network-informed version of the *backfitting* algorithm (Hastie et al. 2017, Chapter 9) for model estimation. Let $\hat{\boldsymbol{\mu}}_{(t_1:t_n)}$, $\hat{\boldsymbol{\pi}}_{(t_1:t_n)}$ denote the fitted evaluated values, and $\boldsymbol{\lambda}_\mu$, $\boldsymbol{\lambda}_\pi$ and $\boldsymbol{\lambda}_X$ denote the hyperparameters of each loss function. The backnet-fitting algorithm iterates the following steps until convergence.

Backnet-fitting algorithm

Given the current fitted values $\boldsymbol{\mu}_{(t_1:t_n)}^{(\ell)}$, $\boldsymbol{\pi}_{(t_1:t_n)}^{(\ell)}$ and coefficients $\boldsymbol{\phi}_{[C]}^{(\ell)}$, do the following:

1. Update the Td -dimensional GNAR term: $\hat{\mathbf{x}} = \mathbf{y} - (\boldsymbol{\mu}_{(1:T)}^{(\ell)} + \boldsymbol{\pi}_{(1:T)}^{(\ell)} + \mathbf{V}\boldsymbol{\phi}_{[C]}^{(\ell)})$, then build the current $nd \times q$ GNAR design $\mathbf{R}(\hat{\mathbf{x}})$ to estimate $\boldsymbol{\theta}_X^{(\ell+1)}$ by (14) or penalised least squares, e.g., (17).
2. Compute $\hat{\mathbf{u}}_\mu = \mathbf{y} - (\boldsymbol{\pi}_{(t_1:t_n)}^{(\ell)} + \mathbf{V}\boldsymbol{\phi}_{[C]}^{(\ell)} + \mathbf{R}(\hat{\mathbf{x}})\boldsymbol{\theta}_X^{(\ell+1)})$, i.e., the nd -dimensional trend residuals resulting from the autoregressive fit in the previous steps, and estimate the $(\ell + 1)$ -step fitted values, e.g., for a smoother matrix $\mathbf{S}_\mu$, these are $\boldsymbol{\mu}_{(t_1:t_n)}^{(\ell+1)} = \mathbf{S}_\mu \hat{\mathbf{u}}_\mu$.
3. Compute $\hat{\mathbf{u}}_\pi = \mathbf{y} - (\boldsymbol{\mu}_{(t_1:t_n)}^{(\ell+1)} + \mathbf{V}\boldsymbol{\phi}_{[C]}^{(\ell)} + \mathbf{R}(\hat{\mathbf{x}})\boldsymbol{\theta}_X^{(\ell+1)})$, i.e., the nd -dimensional seasonal residuals resulting from the autoregressive fit in the previous steps, and estimate the $(\ell + 1)$ -step fitted values, e.g., for a smoother matrix $\mathbf{S}_\pi$, these are $\boldsymbol{\pi}_{(t_1:t_n)}^{(\ell+1)} = \mathbf{S}_\pi \hat{\mathbf{u}}_\pi$.
4. Compute $\hat{\mathbf{u}}_Y = \mathbf{y} - (\boldsymbol{\mu}_{(t_1:t_n)}^{(\ell+1)} + \boldsymbol{\pi}_{(t_1:t_n)}^{(\ell+1)} + \mathbf{R}(\hat{\mathbf{x}})\boldsymbol{\theta}_X^{(\ell+1)})$, i.e., the residuals after removing trend and autoregressive terms fitted in the previous steps, and estimate the $(\ell+1)$ -step fixed effect coefficients $\boldsymbol{\phi}_{[C]}^{(\ell+1)}$ by least squares or generalised least squares.

The design matrices change at every iteration, until convergence, to adapt to the autoregressive unobserved term. This is a major difference from backfitting algorithms for independent data (Wood 2017). See Algorithm 3 in Section D of the supplementary material for details.

can we increase the referencing to Wood?

Proposition 1. *Algorithm 3 solves the convex optimisation problem*

$$\hat{\boldsymbol{\theta}}_{\mathbf{X}}, \hat{\boldsymbol{\phi}}_{[C]} = \underset{\boldsymbol{\theta}_{\mathbf{X}}, \boldsymbol{\phi}_{[C]}}{\operatorname{argmin}} \left\{ \|\mathbf{y} - (\boldsymbol{\mu}_{(t_1:t_n)} + \boldsymbol{\pi}_{(t_1:t_n)} + \mathbf{V}\boldsymbol{\phi}_{[C]} + \mathbf{R}\boldsymbol{\theta}_{\mathbf{X}})\|_2^2 + \sum_{*} \mathcal{J}_*(\boldsymbol{\theta}_*; \boldsymbol{\lambda}_*) \right\}, \quad (19)$$

where $* \in \{\boldsymbol{\mu}, \boldsymbol{\pi}, \mathbf{X}\}$, and $\mathcal{J}_{\boldsymbol{\mu}}$, $\mathcal{J}_{\boldsymbol{\pi}}$ and $\mathcal{J}_{\mathbf{X}}$ are the penalty terms linked to each loss function.

4 Statistical efficiency for GNAR model specifications

Asymptotic results for GNAR models can be obtained by properly reformulating the model into a VAR setting and directly translating results from VAR theory (Hang Yin & Michailidis 2023). However, GNAR models are far more parsimonious than VAR ones. We present nonasymptotic ℓ_2 -norm upper bounds for our conditional least squares estimator given by (14). These elucidate the “blessing of dimensionality” GNAR models have if valid.

Assumptions 1. *Assume that $\mathbf{X}_t \in \mathbb{R}^d$ is a stationary GNAR process given by (8) from which we observe a realisation of length T and $\hat{\boldsymbol{\theta}}$ is given by (14). Hence, we can express the data as $\mathbf{y} - \mathbf{R}\boldsymbol{\theta}_0 = \mathbf{u}$ and $\mathbf{u} \sim \text{WN}(0, \mathbf{I}_n \otimes \boldsymbol{\Sigma}_{\mathbf{u}})$, where $\boldsymbol{\theta}_0 \in \mathbb{R}^q$ is the “true” vector of parameters, and we assume that*

1. *Recall the c -community matrices $\mathbf{R}_c$, which were defined just after (11). The smallest eigenvalues of the matrices $(\mathbf{R}_c^T \mathbf{R}_c)$ are lower bounded by a positive constant, i.e., there exists a $\tau_c > 0$ s.t. $\lambda_{\min}(\mathbf{R}_c^T \mathbf{R}_c) \geq \{\tau_c |K_c| (T - p_c)\}^{1/2}$ for all $c \in [C]$.*
2. *There exists a positive constant Γ such that the ℓ_2 -norm of all columns in $\mathbf{R}$ are upper bounded by $\Gamma/\sqrt{n d^2}$, i.e., there exists a $\Gamma > 0$ s.t. $\max_{j \in [q]} \{\|\mathbf{R}_{\cdot j}\|_2\} \leq \Gamma/\sqrt{n d^2}$, where $n := \min_{c \in [C]} \{(T - p_c)\}$.*
3. *The entries in $\mathbf{u}_t$ are, not necessarily independent, sub-Gaussian white noise, i.e., for each $u_{i,t}$ there exists a $\sigma_i^2 > 0$ such that $E\{\exp(z u_{i,t})\} \leq \exp(z^2 \sigma_i^2 / 2)$ for all $z \in \mathbb{R}$, and $\mathbf{u}_t, \mathbf{u}_s$ are uncorrelated at all times t , i.e., $\text{cov}(\mathbf{u}_t, \mathbf{u}_s) = \mathbb{I}(t = s) \odot \boldsymbol{\Sigma}_{\mathbf{u}}$.*

Assumption 1 is standard in (conditional) least squares settings, it is equivalent to assuming that $\text{rank}(\mathbf{R}) = q \leq d(T - p)$, if satisfied, then $\hat{\boldsymbol{\theta}}$ is the unique least squares solution for (13). If the columns of $\mathbf{R}$ are correlated, then τ should be closer to zero and equal to zero if one column is a linear combination of the other ones, i.e., $\text{rank}(\mathbf{R}) < q$.

Assumption 2 states that no column in $\mathbf{R}$ presents a large variance, i.e., the ℓ_2 -norm can be bounded uniformly across columns. Interestingly, this analysis can be made more precise by considering that the ℓ_2 -norm of each column is the estimated variance of a stationary process. That is, $(\mathbf{R}^T \mathbf{R})$ is related to the autocovariance matrix of the process; see Proposition 2 in Appendix K.3.

Assumption 3 is the most idealised, or at least as idealised as assuming that we observe a stationary process. It, loosely speaking, states that the residuals are stochastically dominated by a normal distribution, hence, outliers (i.e., large deviations from zero) are as unlikely as when sampling from a Gaussian distribution. This condition can be relaxed and studied under sub-exponential, or sub-Weibull assumptions. However, the resulting bounds will not be as tight, and estimation could require more robust methods than least squares. We state our results below.

Theorem 1. Assume that 1 and 2 in Assumptions 1 hold and define the following

- $q_c := \{p_c + (1 + |\mathcal{I}_c|) \sum_{k=1}^{p_c} s_k(c)\}$, i.e., the number of unknown parameters in K_c ,
- $q_{[C]} := \max_{c \in [C]} \{q_c\}$, i.e., the maximum number of unknown parameters,
- $p := \max_{c \in [C]} (p_c)$, i.e., the largest active lag across all communities,
- $d_{(1)} := \min_{c \in [C]} \{|K_c|\}$, i.e., the size of the smallest community,
- $\tau := \min_{c \in [C]} (\tau_c)$, i.e., the largest constant for uniformly lower bounding the smallest eigenvalues of all $(\mathbf{R}_c^T \mathbf{R}_c)$ (i.e., $\lambda_{\min}(\mathbf{R}_c^T \mathbf{R}_c) \geq \{\tau d_{(1)}(T - p)\}^{1/2}$ for all $c \in [C]$).

Then, we have that

$$\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|_2 \leq 2\{\tau d_{(1)}(T - p)\}^{-1/2} (Cq_{[C]})^{1/2} \|\mathbf{R}^T \mathbf{u}\|_\infty, \quad (20)$$

further, if we assume that A.3 holds, then for any $\delta > 0$ we also have that

$$\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|_2 \leq 2(Cq_{[C]})^{1/2} \sigma_{\mathbf{u}} \Gamma \sqrt{2} \left[\left\{ \log \left(\sum_{c=1}^C q_c \right) \{\tau d_{(1)}(T - p)\}^{-1} \right\}^{1/2} + \delta \right], \quad (21)$$

holds with probability at least

$$1 - 2 \exp \left[-\delta^2 \{\tau d_{(1)}(T - p)\} \right], \quad (22)$$

where $\sigma_{\mathbf{u}}^2 = \max_{i \in \mathcal{K}} (\sigma_i^2)$ is the largest variance among nodal white noise processes $u_{i,t}$.

Theorem 1 highlights the strengths and weaknesses of GNAR models. For the bound in (20) to be meaningful, it is necessary that no community is empty, and that the linear model obtained by conditioning the realisation is not underdetermined (i.e., $d_{(1)} > 0$ and $\tau > 0$). Also, as the number of communities and/or interactions increases the bound becomes looser. That is, larger q result in a looser bound for a given realisation $\{\mathbf{X}_t\}$ of length T . A small τ also results in a looser bound, i.e., the more correlated the columns of $\mathbf{R}$ are, the looser the bound is. Thus, the autocorrelation structure that the network induces (i.e., $\mathbf{W}_t$ and $\mathbf{S}_r$) affect how small τ is and how tight (20) is for a given network time-series, and how quickly the probability given by (22) approaches one. The scaling in Theorem 1 is expected for linear models; see Portnoy (1984).

Interestingly, the ratio, $\{Cq_{[C]} \log(Cq_{[C]})\} / \{\tau d_{(1)}(T - p)\}$, between problem and sample size becomes smaller as GNAR models become more parsimonious, which by Theorem 1 makes estimation more efficient and precise. Thus, Theorem 1 shows the increase in statistical efficiency as the model becomes more parsimonious. Whereas, by (21), as GNAR models become less parsimonious, then the bounds become looser. Hence, if a GNAR model is valid, then increasing dimension can be an asset when performing estimation, which enables modelling of very *high-dimensional* (network) time series. By Theorem 1, we obtain the following result.

Corollary 1. *Assume that A.1, A.2 and A.3 in Assumptions 1 hold. Then, the ratio between problem size and observations goes to zero*

$$\lim_{T \rightarrow \infty} \{Cq_{[C]} \log(Cq_{[C]})\} \{\tau d_{(1)}(T - p)\}^{-1} = 0, \text{ a.s.,}$$

and for all $\delta > 0$ we have that

$$\lim_{T \rightarrow \infty} \mathbf{Pr} \left(\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|_2 > \delta \right) = 0. \quad (23)$$

Corollary 1 tells us that the least squares estimator (14) is consistent, whether the network time series is invertible or not. Further, since $\hat{\boldsymbol{\theta}}$ is a consistent *M-estimator*, we also have the following asymptotic limit result.

Theorem 2. *Assume that A.1, A.2 and A.3 in Assumptions 1 hold. Then, we have that*

$$\sqrt{d_{(1)}(T - p)} \left(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0 \right) \xrightarrow{d} N_q(0, \boldsymbol{\Sigma}_{\hat{\boldsymbol{\theta}}}), \quad (24)$$

that is, $(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$ is zero-mean asymptotically normal with covariance

$$\boldsymbol{\Sigma}_{\hat{\boldsymbol{\theta}}} = (1/d) \{E(\mathbf{R}_t^T \mathbf{R}_t)\}^{-1} \{E(\mathbf{R}_t^T \boldsymbol{\Sigma}_u \mathbf{R}_t)\} \{E(\mathbf{R}_t^T \mathbf{R}_t)\}^{-1},$$

and that the empirical covariance estimator $\hat{\boldsymbol{\Sigma}}_u$,

$$\hat{\boldsymbol{\Sigma}}_u = \frac{1}{T - p} \sum_{t_i=p+1}^T (\mathbf{y}_{t_i} - \mathbf{R}_{t_i} \hat{\boldsymbol{\theta}})(\mathbf{y}_{t_i} - \mathbf{R}_{t_i} \hat{\boldsymbol{\theta}})^T,$$

is a consistent estimator of $\boldsymbol{\Sigma}_u$.

Theorem 1 and Corollary 1 highlight that our GNAR framework provides an easily generalisable family of linear models. For example, extending (12) to generalised linear (McCullagh & Nelder 1989) or state-space models (Brockwell & Davis 2006, Chapter 10). Note that $\hat{\boldsymbol{\theta}}$ given by (14) is consistent even if the white noise processes are correlated. Interestingly, the distinction between using the generalised least squares estimator and the least squares ones is evidenced by looking at the tighter bounds in (21) and asymptotic covariance in (24). As expected, the generalised least squares estimator in (15) results in tighter bounds and a more efficient asymptotic covariance. See Appendix J and K.3 in the supplement for a proof of Theorem 1 and Corollary 1.

Remark 3. *Assume that $\text{cov}(\mathbf{u}_t) = \boldsymbol{\Sigma}_u$ and $\hat{\boldsymbol{\theta}}_{\text{gls}}$ is given by (15). Then, the bound in (21) becomes tighter by a $\sqrt{d}$ -scaling, that is,*

$$\sqrt{d} \|\hat{\boldsymbol{\theta}}_{\text{gls}} - \boldsymbol{\theta}_0\|_2 \leq 2(Cq_{[C]})^{1/2} \sigma_u \Gamma \sqrt{2} \left[\{\log(q) \{\tau d_{(1)}(T - p)\}^{-1}\}^{1/2} + \delta \right],$$

and $\boldsymbol{\Sigma}_{\hat{\boldsymbol{\theta}}}$ in (24) is given as

$$\boldsymbol{\Sigma}_{\hat{\boldsymbol{\theta}}} = (\sigma_u^2 / |\mathcal{K}|) \{E(\mathbf{R}_t^T \mathbf{R}_t)\}^{-1}, \quad (25)$$

where there exist $\tau_c > 0$ such that $\lambda_{\min}(\mathbf{R}_c^T \boldsymbol{\Sigma}_T^{-1} \mathbf{R}_c) \geq \{\tau_c |K_c| (T - p_c)\}^{1/2}$ for all $c \in [C]$

Our results generalise known-ones for linear models (Portnoy 1985) to the network time series setting. The highly *parsimonious* GNAR framework enables us to study the unknown parameters as *vectors* rather than as *matrices* and seeing ‘dimension as an asset’ (as mentioned above).

4.1 Estimation theory for dynamic GNAR structural models

Due to space constraints we do not explore alternative estimators, nor do we derive detailed results for different choices of loss functions. However, we state general conditions for consistent estimation and inference for $\phi_{[C]}$ in Theorem 3.

Assumptions 2. Assume that the data are a realisation of length T generated by (4), and that our choice of estimators and the unknown “true” functions satisfy the following.

1. For all $i \in [d]$, there exists a square-integrable function $\mathbf{h}_i : \mathbb{R} \rightarrow \mathbb{R}^{q_v}$ such that

$$\mathbf{h}_i(t) = \mathbb{E}(\mathbf{v}_{i,t}), \text{ where } \mathbf{h}_i(t) = \{h_{i,1}(t), \dots, h_{i,q_v}(t)\}, \text{ and} \\ \sum_{k=1}^{q_v} \int_0^T \mu_i(t) h_{i,k}(t) dt = 0, \sum_{k=1}^{q_v} \int_0^T \pi_i(t) h_{i,k}(t) dt = 0, \sum_{k=1}^{q_v} \int_0^T h_{i,k}''(t) dt < \infty. \quad (\text{A.4})$$

2. For each $c \in [C]$, we have that the asymptotic biases vanish at the following rates

$$\{|K_c| (T - p_c)\}^{1/2} \left\{ \frac{1}{\sqrt{|K_c|}} \sum_{i \in K_c} \|\hat{\mu}_i - \mu_i\|_{L_2}^2 \right\} = o_{\mathbf{Pr}}(1), \\ \{|K_c| (T - p_c)\}^{1/2} \left\{ \frac{1}{\sqrt{|K_c|}} \sum_{i \in K_c} \|\hat{\pi}_i - \pi_i\|_{L_2}^2 \right\} = o_{\mathbf{Pr}}(1). \quad (\text{A.5})$$

Assumption (A.4) says that, for each node i , the time-varying predictors $\mathbf{v}_{i,t}$ have a smooth-enough trend, which could be a constant (e.g., the node-wise mean $\bar{\mathbf{v}}_i$), and that these are not redundant temporal or seasonal terms. That is, $\mathbf{v}_{i,t} - \mathbf{h}_i(t) \neq 0$ for all i and t , and $\mathbf{h}_i$ is orthogonal to the community-wise sum of temporal and seasonal trends in (3), which ensures identifiability. Intuitively, this condition says that the predictors are not a temporal or seasonal term, but rather another variable that has an *identifiable* effect on the mean for each node i in K_c . In other words, these explain the remaining variation in $Y_{i,t}$ after removing temporal and seasonal trends, *and* accounting for serial correlation.

Assumption (A.5) says that for each $c \in [C]$ the sum of temporal and seasonal biases vanishes at the same asymptotic rate as $\hat{\phi}_c$ converges in probability to ϕ_c . This does not require all node-wise biases to vanish at the same rate. Instead, the impact of “misbehaved” nodes is controlled by “averaging” out the bias as a community-wise mean. Intuitively, (A.5) says that the bias does not dominate the community-wise standard error, which allows us to derive the asymptotic distribution of our estimator. Both assumptions generalise accepted ones for *univariate* partial linear regression (Gannaz 2007) to longitudinal networks. However, these can be relaxed at the cost of the $\{|K_c| (T - p_c)\}^{1/2}$ rate.

Theorem 3. Assume that (A.4) and (A.5) hold, and that 1, 2 and 3 in Assumptions 1 are valid. Then, $\hat{\phi}_{[C]}$ given by (19) is consistent and asymptotically normal, that is,

$$\sqrt{d_{(1)} (T - p)} \left(\hat{\phi}_{[C]} - \phi_{[C]} \right) \xrightarrow{d} N_{q_v}(0, \Sigma_{\hat{\phi}}), \text{ where} \\ \Sigma_{\hat{\phi}} = \{\mathbb{E}(\tilde{\mathbf{V}}_t^T \tilde{\mathbf{V}}_t)\}^{-1} \{\mathbb{E}(\mathbf{V}_t^T \Sigma_{\mathbf{u}} \mathbf{V}_t)\} \{\mathbb{E}(\tilde{\mathbf{V}}_t^T \tilde{\mathbf{V}}_t)\}^{-1}, \quad (26)$$

$\tilde{\mathbf{V}}_t = \mathbf{V}_t - \mathbf{H}(t)$, and the i th row in $\mathbf{H}(t)$ is $\mathbf{h}_i^T(t)$. Further, the asymptotic distributions of $\hat{\theta}_{\mathbf{X}}$ in (14) and (15) are still given by (24) and (25).

See Appendix L for a proof of Theorem 3 and Proposition 1. Under commonly accepted regularity conditions there are well-known nonparametric consistent estimators (e.g., smoothing splines) for univariate functions, as required by Theorem 3.

Is this remark in lieu of theoretical results on the μ and π ?

4.2 Hyperparameter tuning and forward-validation

We suggest exploiting the additive model representation in (12) or (4) to easily express a realisation in different blocks and forward-validation (cross-validation using subsequent time blocks to account for serial correlation) for hyperparameter tuning and selection; see Hjorth (1982), Cerqueira et al. (2017). We intend to pursue these issues in more detail in future work, and derive error bounds like the one in (20) for each community estimator.

So far we have used the network as hard fixed prior information, which we believe is sensible and has proven useful in many scenarios; see, e.g., Nason et al. (2025). However, we can augment Algorithm 3 and include network estimation based on a data-driven criterion. As suggested by Nason & Palasciano (2025), we propose generating random networks and selecting the one that minimises the one-step ahead forecast squared error for a fixed GNAR model order and choice of nonparametric estimators. Algorithm 2 incorporates network estimation into Algorithm 3.

4.3 Model order selection

Consider a model order selection criterion given by

$$\text{Cr}(\mathbf{y}, \mathbf{R}; \hat{\boldsymbol{\theta}}) := \log |\hat{\boldsymbol{\Sigma}}_{\mathbf{u}}| + \frac{1}{n} \sum_{c=1}^C \|\hat{\boldsymbol{\theta}}_c\|_{\text{comp}}, \quad (27)$$

where $\hat{\boldsymbol{\theta}}_c$ are given by (14) and, in an abuse of notation, $\|\cdot\|_{\text{comp}}$ denotes model order complexity for community K_c . If we assume that $\mathbf{u}_t \sim \text{N}(0, \boldsymbol{\Sigma}_{\mathbf{u}})$, then the first term in (27) corresponds to the log-likelihood evaluated at its maximum. Further, if we set $\|\hat{\boldsymbol{\theta}}_c\|_{\text{comp}} = 2q_c$ then (27) is equal to the AIC (Akaike 1973), if $\|\hat{\boldsymbol{\theta}}_c\|_{\text{comp}} = q_c \log(n_c)$, then (27) equals the BIC (Schwarz 1978), and if $\|\hat{\boldsymbol{\theta}}_c\|_{\text{comp}} = 2q_c \log\{\log(n_c)\}$, then (27) equals the HQC (Hannan & Quinn 1979).

Lemma 1. *Assume that $\mathbf{X}_t$ is a stationary GNAR process given by (12), and that $\hat{\boldsymbol{\theta}}$ is given by (14). Then, the criteria, BIC and HQC, given by (27) are lag-order consistent,*

$$\lim_{T \rightarrow \infty} \mathbf{Pr}(\hat{p} = p_0) = 1, \quad (28)$$

where $\hat{p}$ is model order suggested by BIC or HQC and p_0 is the unknown “true” model order. Further, for AIC we have that

$$\lim_{T \rightarrow \infty} \mathbf{Pr}(\hat{p} < p_0) = 0. \quad (29)$$

If $\hat{\boldsymbol{\Sigma}}_{\mathbf{u}}$ is block-diagonal, then $\log |\hat{\boldsymbol{\Sigma}}_{\mathbf{u}}| = \sum_{c=1}^C \log |\hat{\boldsymbol{\Sigma}}_{\mathbf{u},c}|$, and we can minimise our chosen criterion in a community-wise fashion. Alternatively, we can use the R-Corbit plot proposed by Nason et al. (2025) to quickly determine model order for each community.

5 Simulation study

We study the finite-sample properties of our estimators and model selection criteria as follows. We generate five realisations of different lengths from (12) and compute the three criteria in (27), and repeat the experiment one-hundred times to assess the effectiveness. We further simulate high-dimensional longitudinal network data and evaluate the performance of Algorithm 5 by repeating one-step ahead forecasts. We conclude our simulation studies by generating dynamic GNAR realisations from (4) and study the properties of our fixed effects estimator. The results are in Table AA. See Section in the Supplement for details.

6 Applications to geology and political science

Describe briefly and quickly both applications.

6.1 Longitudinal network analysis of Brumandhino data

6.2 Longitudinal study of presidential elections in the USA

We study the twelve presidential elections in the USA from 1976 to 2020. The data are obtained from the MIT Election Data and Science Lab (doi.org/10.7910/DVN/42MVDX). We model the longitudinal network $(\mathbf{X}_t, \mathcal{G})$, where $X_{i,t}$ is the percentage of votes for the Republican nominee in the i th state (ordered alphabetically) for election year $t \in \{1976, 1980, \dots, 2020\}$ as a community- α GNAR model. The underlying network is $\mathcal{G} = (\mathcal{K}, \mathcal{E})$, where $i \in \{1, \dots, 51\}$ and $d = 51$, and it is built by connecting states that share a land border. Alaska and Hawaii are not connected to the network, however, each one belongs to its respective community and is related to other states by sharing the communal α coefficients. Following Shin & Webber (2014), Ishise & Matsuo (2015), we classify states into communities based on the percentage of elections won by either party, i.e., each node (state) is classified as either *Red*, *Blue* or *Swing*, which are identified by the covariates $c \in \{1, 2, 3\}$. The communities are: $i \in K_1$ if the Republican nominee won at least 75% of elections, $i \in K_2$ if the Democrat nominee won at least 75% of elections, and $i \in K_3$ if neither nominee won at least 75% of elections; see Figure 2.

Def refer to Costa Rica paper, and shorten this?

6.2.1 Analysing interactions between communities

We also analyse if there are differences in magnitude for interaction coefficients, and how these states vote after a particular result, e.g., do *Blue* states increase their vote for a Democrat nominee more than *Swing* states do after a Republican win? Modelling the presidential election time series as a community- α GNAR model with interactions enables us to shed light on these questions. Based on the network autocorrelation in Section A.1, we select the following model orders: $(2, [1, 1], \{2, 3\})$ for *Red* states, $(2, [1, 1], \{1, 3\})$ for *Blue* and $(2, [1, 1], \{1, 2\})$ for *Swing*. Table 1 shows the lag two estimated interaction coefficients. Interestingly, the interaction coefficient estimates for the influence of *Blue* and *Red* states on *Swing* states ($\gamma_{2,1,3:1}, \gamma_{2,1,3:2}$) are smaller in magnitude than those for the influence of *Swing* states on *Blue* and *Red* states. Further, $\hat{\gamma}_{2,1,2:1} = -1.14$ (effect of *Red* states on *Blue* ones), suggests that *Blue* states vote as strongly for the Democrat nominee as *Red* states

Table 1: Estimated lag-two interaction coefficients $\hat{\gamma}$ for the standardised network time series of vote percentages for the Republican nominee in presidential elections in the USA from 1980 to 2020. The fitted model is a community- α GNAR model with interactions among *Blue*, *Red* and *Swing* states. Model order is GNAR([2, 2, 2]; {[1, 1], [1, 1], [1, 1]}; [3]; {2, 3}, {1, 3}, {1, 2}). [Recall from (7) that the interaction coefficients are $\gamma_{k,r,c;\tilde{c}}$, where k is lag, r is network r -stage and $c : \tilde{c}$ are two different communities.]

	$\hat{\gamma}_{2,1,1:2}$	$\hat{\gamma}_{2,1,1:3}$	$\hat{\gamma}_{2,1,2:1}$	$\hat{\gamma}_{2,1,2:3}$	$\hat{\gamma}_{2,1,3:1}$	$\hat{\gamma}_{2,1,3:2}$
Estimate	-0.480	-0.490	-1.140	-0.785	-0.300	-0.169
Std. Err.	0.396	0.214	0.332	0.219	0.246	0.211

voted for the Republican nominee eight-years before. Whereas, *Red* states react half as strongly ($\hat{\gamma}_{2,1,1:2} = -0.48$) to *Blue* states. This suggests that voters in *Blue* states tend to flock (vote strongly for a Democrat nominee), whilst, *Red* states do not react as strongly. Moreover, *Swing* states are influenced less strongly by votes from *Red* and *Blue*, which suggests these states might be less polarised. Remarkably, *Blue* states have the largest polarisation as measured by the estimated interaction coefficient magnitude (i.e., $|\hat{\gamma}_{2,1,2:1}|$).

6.3 Discussion

We use the community- α with interactions GNAR model for analysing the effects within and between state-wise communities in the USA during presidential elections from 1976 to 2020. Our main interest is identifying noteworthy differences in the voting patterns among each community. However, it is worth mentioning that, as expected, *Red* states vote mostly for the Republican nominee, *Blue* states for the Democrat nominee, and that *Swing* states play the deciding role, given that the community-wise mean percentage of votes for the Republican nominee is 50.80%. However, somewhat surprisingly, based on the global- α GNAR model forecasting results, our analysis suggests that communities behave more similarly than expected, i.e., there appear to be (spatio-temporal) global and communal network effects. Further, *Blue* states vote in a block-wise manner, and have the strongest reaction (interaction coefficient magnitude) by increasing the number of votes for a Democrat nominee with respect to previous elections. *Red* and *Swing* states react at 50% and at 25% the strength of *Blue* states. Our autocorrelation analysis suggests that alternating between a Democrat and Republican president every eight-years might not be the only governing dynamic. Following the 2016 election, the new pattern could be that parties take turns at the presidency every four rather than eight years. Interestingly, our results corroborate the analysis of political polarisation in Twitter and Reddit longitudinal networks in Xiaojing et al. (2023). We highlight that the observed data is overly sparse, thus, inferences and forecasts should be treated with caution, and that these claims require further investigation.

Put simply, the parsimonious GNAR model allows us to investigate the dynamics of presidential election cycles in the USA. Our analysis highlights the eight-year long presidential cycle, a possible change in the system after the 2016 election, the deciding role *Swing* states have, and behaviour differences between *Blue* and *Red* states.

Conclusion

We presented a community- α GNAR model with different lag and r -stage orders for different communities and inter-community interactions, which are not constrained to be symmetric. Such models can detect differences in the dynamics of communities present in a network time series. Further, their parsimonious framework allows us to analyse high-dimensional data with reasonable length time series, such as the presidential election data studied in Section 6.2. However, it is important to be clear about its limitations: our models require knowledge of the number and membership of communities and assume that the network time series is stationary. Future work will focus on relaxing these limitations and developing a clustering algorithm aimed at exploiting the community-wise correlation structure in the network time series by assigning each node to the community that minimises some goodness-of-fit criterion based on the network autocorrelation function (NACF).

The code for reproducing our results and fitting community- α GNAR models is available at [githublink \(to add\)](#).

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Supplementary Material

- **Appendix B:** Notation summary.
- **Appendix E:** Outline of fused group lasso clustering method.
- **Appendix G:** GNAR simulations and plot of finite-sample error bounds.
- **Appendix H:** Analysis of one-lag differenced presidential election data.
- **Appendix I:** Proof of stationarity conditions for GNAR models.
- **Appendix J:** Proof of Theorem 1 (finite-sample error bounds).
- **Appendix K:** Proof of Corollary 1 (asymptotic normality and consistency).
- **GitHub link:** https://github.com/dansal182/community_GNAR.

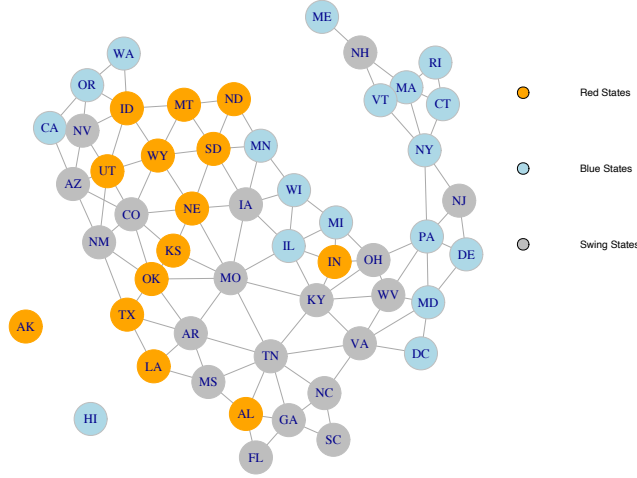


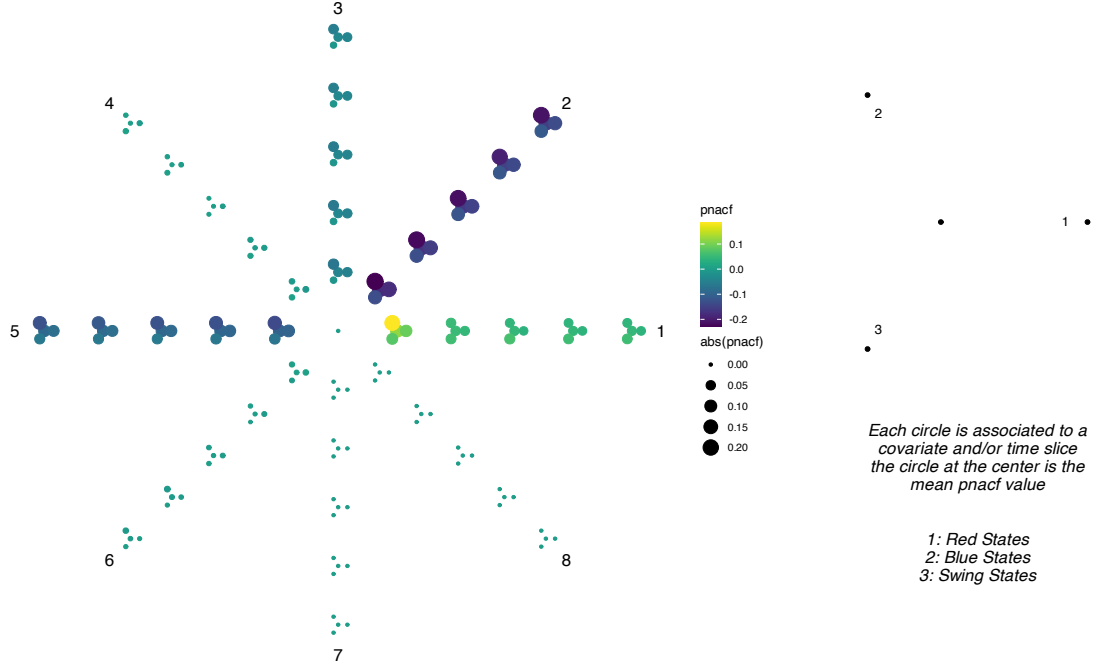
Figure 2: USA state-wise network, blue nodes are *Blue* states (Democrat nominee won at least 75% of elections), orange nodes are *Red* states (Republican nominee won at least 75% of elections), and grey nodes are *Swing* states (neither party won at least 75% of elections).

A Details for Section 6.2

A.1 Analysis of network autocorrelation and model selection

The R-Corbit plot in Figure 3 shows that the PNACF is positive at the first lag, negative and strongest at the second lag, considerably smaller at lags three and four, and interestingly, appears to be strong at the fifth lag across all r -stages, and cuts off after the fifth. At the first lag, the PNACF is visually smaller after the first r -stage. At both the second and fifth lags, the PNACF actually decays as r -stage grows, but is not easy to see from the plot. It does not cut-off at any r -stage. The positive correlation at lag one corresponds to elections in which a president is running for reelection, and that network affects influence said election. Remarkably, the strong negative correlation at the second lag across all r -stages suggests a change in the system, which we interpret as alternating between Republican and Democrat nominees once the incumbent president has completed their eight-year term. This has been the case with the exception of Jimmy Carter (1976–1980), George Bush (1988–1992) and Donald J. Trump (2016–2020). Interestingly, the exception cases correspond to elections where an incumbent president was running for reelection (i.e., half-terms). We believe that the fifth lag might be identifying with these exception cases.

Based on the PNACF analysis in Figure 3, and that the lag five PNACF values are smaller than the lag two ones, and given that we only have data from twelve elections, we initially compare models of order two. Hence, we trial a community- α GNAR(2, $\{[1, 0]\}$, 3) model. The parsimonious nature of GNAR models allows us to estimate multiple models and analyse autocorrelation up to lag eight, in spite of having twelve observations for a vector of dimension fifty-one.



pnacf R-Corbit plot with 3 time frames or samples, max lag and max path length 5

Figure 3: PNACF R-Corbit plot of the series $\mathbf{X}_t$. The underlying network is the USA state-wise network; see Figure 2, and Section 3 in Nason et al. (2025) for plot description.

Table 2: Estimated coefficients for the standardised network time series of vote percentages for the Republican nominee in presidential elections in the USA from 1976 to 2020. The fit is a community- α GNAR(2, $\{[1, 0]\}$, 3). [Recall from (8) that the coefficients are α_{kc} and β_{krc} , where k is lag, r is network r -stage and c is community.]

	$\hat{\alpha}_{11}$	$\hat{\beta}_{111}$	$\hat{\alpha}_{21}$	$\hat{\alpha}_{12}$	$\hat{\beta}_{112}$	$\hat{\alpha}_{22}$	$\hat{\alpha}_{13}$	$\hat{\beta}_{113}$	$\hat{\alpha}_{23}$
Estimate	0.393	0.183	-0.593	0.558	0.069	-0.351	0.905	-0.747	-0.591
Std. Err.	0.104	0.200	0.066	0.108	0.178	0.063	0.089	0.162	0.061

Table 2 shows that if (24) is valid, then a *Swing* states network effect null hypothesis, $H_0 : \beta_{113} = 0$, is rejected and statistically significant at the 5% level (the t -value is -5.05 , giving a p -value < 0.001), which suggests strong inter-state communication within the *Swing* state community. The network effect for *Red* and *Blue* states do not reject the null hypothesis of zero for β_{111}, β_{112} , even though the former's estimate is of a reasonable size.

A.2 Comparison with alternative models

Table 3 compares our choice with alternative models. Before model fitting we centre and standardise the data to bring it closer to our modelling assumptions (*homoscedastic* noise and zero-mean). That is, we compare the model's forecasting performance using the standardised time series $Y_{i,t} := \sqrt{11} \{ \sum_{t=1}^{11} (X_{i,t} - \bar{X}_i)^2 \}^{-1/2} (X_{i,t} - \bar{X}_i)$, where $\bar{X}_i = (1/11) \sum_{t=1}^{11} X_{i,t}$. Each model produces a one-step ahead standardised forecast $\hat{Y}_{i,12}$, which we use to compute the one-step ahead forecast as $\hat{X}_{y_i,12} = \{ \sum_{t=1}^{11} (X_{i,t} -$

Table 3: Model comparison of root mean-squared prediction error (RMSPE). GNAR... models increase in complexity from global (glo), to community (com), to interactions (int), to local (loc). Sp. VAR: sparse VAR(2). CARar: Conditional temporal autoregressive (spatially-informed noise covariance). $\#||\boldsymbol{\theta}||_0$ is number of non-zero parameters.

	GNAR _{glo}	GNAR _{com}	GNAR _{int}	GNAR _{loc}	Sp. VAR	CARar
RMSPE($\hat{\mathbf{X}}_{12}$)	2.34	2.94	3.53	24.95	72.57	3.60
RMSPE($\hat{\mathbf{Y}}_{12}$)	3.32	4.44	3.67	5.22	4.07	2.75
$\# \boldsymbol{\theta} _0$	6	12	42	106	176	6(+51)

$\bar{X}_i)^2\}^{1/2} \hat{Y}_{i,12}/\sqrt{11} + \bar{X}_i$. For completeness, we also fit models to the observed time series $\{\mathbf{X}_t\}$ (not standardising) and analyse the root mean-squared prediction error (RMSPE) for model output $\hat{\mathbf{X}}_t$, i.e., for the $\mathbf{Y}_t$, $\text{RMSPE}(\hat{\mathbf{Y}}_{12}) = \{\sum_{i=1}^{51} (X_{i,12} - \hat{X}_{y_i,12})^2/51\}^{1/2}$, and for $\hat{\mathbf{X}}_t$, $\text{RMSPE}(\hat{\mathbf{X}}_{12}) = \{\sum_{i=1}^{51} (X_{i,12} - \hat{X}_{i,12})^2/51\}^{1/2}$. For reproducibility, we fix the random seed, i.e., `set.seed(2024)`, before model fitting; see supplementary material for details. We fit sparse VAR(2) using `sparsevar`; see Vazzoler (2021), and CARar(2), where forecasts are computed as a global- α GNAR as suggested by Nason et al. (2025), using `CARBayesST`; see Lee et al. (2018). We remark that CARar forecasts are equivalent to global- α GNAR ones if estimation is done by generalised least-squares with a spatially-informed covariance matrix; see Nason et al. (2025).

Remarkably, the global- α GNAR(2, [1, 0]) model produces the smallest RMSPE, and is the only model that *outperforms* the Naive (previous observation) forecast, which has a RMSPE equal to **2.45**. This could be related to the small sample error bounds in Theorem 1, since the highly parsimonious global- α GNAR model can be estimated precisely, even in (ultra) high-dimensional settings. We see that parsimonious GNAR models produce accurate forecasts with a limited amount of data. Further, all GNAR models perform comparatively well with standardised and non-standardised data, whereas, not standardising the data severely impacts the performance of sparse VAR.

B Notation summary

- T : length of realisation.
- $\mathcal{X} := (\mathbf{X}_t, \mathcal{G})$: Network time series.
- $X_{i,t}$: Nodal time series.
- $\mathbf{X}_t$: Multivariate time series at time t .
- $\mathcal{G} = (\mathcal{K}, \mathcal{E})$, where $\mathcal{K} = \{1, \dots, d\}$ is the node set, $\mathcal{E} \subseteq \mathcal{K} \times \mathcal{K}$ is the edge set.
- $\mathbf{S}_r \in \{0, 1\}^{d \times d}$, where $[\mathbf{S}_r]_{ij} = \mathbb{I}\{d(i, j) = r\}$, $\mathbb{I}$ is the indicator function, $r \in \{1, \dots, r_{\max}\}$, and $r_{\max} \in \mathbb{Z}^+$ is the longest shortest path in $\mathcal{G}$.
- Weights matrix $\mathbf{W}_t \in \mathbb{R}^{d \times d}$ as the matrix $[\mathbf{W}_t]_{ij} = w_{ij}(t)$.
- $\mathbf{Z}_t^r := (\mathbf{W}_t \odot \mathbf{S}_r) \mathbf{X}_t$, where $\odot$ denotes the Hadamard (component-wise) product.

- $K_c := \{i \in \mathcal{K} : X_{i,t} \text{ is associated with covariate } c\}$.
- $\mathcal{I}_c := \{\tilde{c} \in [C] : c \text{ interacts with } \tilde{c}, \text{ where } \tilde{c} \neq c\}$.
- $\mathbf{Z}_{t-k}^{r,c} := (\mathbf{W}_{t,c} \odot \mathbf{S}_r) \mathbf{X}_{t-k}^c$, r -stage neighbourhood regressions for community K_c .
- $[\mathbf{W}_{t,c}]_{ij} = w_{ij}(t)\mathbb{I}(i \in K_c \text{ and } j \in K_c)$.
- $\alpha_{k,c}$: autoregressive coefficient at lag k for community K_c .
- $\beta_{k,r,c}$: Within community coefficient at lag k for community K_c .
- $\gamma_{k,c,c'}$: Between community coefficient at lag k for the effect community $K_{c'}$ has on community K_c .
- $p_c \in \mathbb{Z}^+$ and $s_k(c) \in \{1, \dots, r_c^*\}$, where the former is the maximum lag for community K_c and the latter is the maximum r -stage depth.
- $\mathbf{Z}_{t-k}^{r,c;\tilde{c}} = \xi_c \odot (\mathbf{W}_t \odot \mathbf{S}_r) \mathbf{X}_{t-k}^{\tilde{c}}$, between community interactions.
- $\sigma_{\mathbf{u}}^2$: white noise process variance.
- $\mathbf{R}_{k,c} = \left[\mathbf{X}_{t-k}^c | \mathbf{Z}_{t-k}^{1,c} | \dots | \mathbf{Z}_{t-k}^{s_k(c),c} | \mathbf{Z}_{t-k}^{1,c;\tilde{c}} | \dots | \mathbf{Z}_{t-k}^{s_k(c),c;\tilde{c}} \right]$.
- $\mathbf{R}_c = [\mathbf{R}_{1,c} | \dots | \mathbf{R}_{p_c,c}]$.
- $\boldsymbol{\theta}_c := (\alpha_{k,c}, \beta_{k,c}, \gamma_{k,c})$.
- $\boldsymbol{\theta} := (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_C)$.
- $\mathbf{y} := (\mathbf{X}_{p+1}, \dots, \mathbf{X}_T)$.
- $\mathbf{y} = \mathbf{R}\boldsymbol{\theta} + \mathbf{u}$: linear model representation.
- $q_c := \{p_c + (1 + |\mathcal{I}_c|) \sum_{k=1}^{p_c} s_k(c)\}$, i.e., the number of unknown parameters in K_c ,
- $q_{[C]} := \max_{c \in [C]} \{q_c\}$, i.e., the maximum number of unknown parameters,
- $p := \max_{c \in [C]} (p_c)$, i.e., the largest active lag across all communities,
- $k_{[C]} := \min_{c \in [C]} \{|K_c|\}$, i.e., the size of the smallest community,
- $\tau := \min_{c \in [C]} (\tau_c)$, i.e., the largest constant for uniformly lower bounding the smallest eigenvalues of all $(\mathbf{R}_c^T \mathbf{R}_c)$ (i.e., $\lambda_{\min}(\mathbf{R}_c^T \mathbf{R}_c) \geq \{\tau k_{[C]}(T - p)\}^{1/2}$ for all $c \in [C]$).

The linear model representation can be exploited to express the problem in general form as follows; see Section K.3 in the supplementary material.

$$\begin{aligned}
\mathbf{R}_{k,t,c} &:= \left[\mathbf{X}_{t-k}^c | \mathbf{Z}_{t-k}^{1,c} | \dots | \mathbf{Z}_{t-k}^{s_k(c),c} | \mathbf{Z}_{t-k}^{1,c;\tilde{c}} | \dots | \mathbf{Z}_{t-k}^{s_k(c),c;\tilde{c}} \right], \\
\mathbf{R}_{k,t} &:= [\mathbf{R}_{k,t,c} | \dots | \mathbf{R}_{k,t,C}], \\
\mathbf{Z}_{t_i} &:= [\mathbf{R}_{1,t} | \dots | \mathbf{R}_{p,t}],
\end{aligned}$$

where each predictor column for $\tilde{c} \in \mathcal{I}_c$ is concatenated in ascending order with respect to $\tilde{c}$, i.e., if $\tilde{c}_2 > \tilde{c}_1$, then the column for $\tilde{c}_1$ precedes the one for $\tilde{c}_2$, $\mathbf{u}_{t_i}$ is a sub-Gaussian multivariate white noise process with a diagonal covariance, and $\boldsymbol{\theta}_0 \in \boldsymbol{\Theta} \subset \mathbb{R}^q$ is the “true” vector of parameters.

C Conditions for stationarity

Remark 4. Assume that $\mathbf{X}_t \in \mathbb{R}^d$ is a particular community- α GNAR process with the same lag order for all communities, that the underlying network is fully connected and static, and that each node i is a community (i.e., $K_c = \{i\}$ for $i \in \{1, \dots, d\}$, where $C = d$). Then, this community- α GNAR model can satisfy one of the following:

- (a) If we include all community interactions for all nodes and assume that r -stage order includes all possible r -stage neighbours at all lags for all nodes (i.e., $\gamma_{k,r,c;\tilde{c}} \neq 0$ for all $r \leq r_{\max}$ at all $k = 1, \dots, p$), then the model given by (8) is identical to an unconstrained VAR(p).
- (b) If we constrain community interactions to a single effect for all lag and r -stage pairs (k, r) , i.e., $\gamma_{k,r,c;\tilde{c}} = \beta_{kr}$ for all c and $\tilde{c}$ at all lags and r -stages, then the model given by (8) is identical to a local- α GNAR($p, [s_k]$). (Note, where each single node i is a community, then there are no other nodes in their community, so there are only interaction effects, γ , but for single effects these become β s.)
- (c) If we constrain community interactions and local effects for all nodes at all lags (i.e., $\alpha_{k,c} = \alpha_k$ and $\gamma_{k,r,c;\tilde{c}} = \beta_{kr}$ for all c and $\tilde{c}$ at all lags and r -stages), then the model given by (8) is identical to a global- α GNAR($p, [s_k]$).

Our model can be thought of as a sum of C global- α GNAR models, so stationarity conditions can be immediately derived from results of Knight et al. (2020).

Theorem 4. [Knight et al. (2020)] Let $\mathbf{X}_t$ be a local- α GNAR($p, [s_k]$) process with associated static network $\mathcal{G} = (\mathcal{K}, \mathcal{E})$. Further, assume that the parameters in

$$X_{i,t} = \sum_{k=1}^p \left(\alpha_{i,k} X_{i,t-k} + \sum_{r=1}^{s_k} \sum_{c=1}^C \beta_{kr,c} Z_{i,t-k}^{r,c} \right) + u_{i,t}, \quad (\text{C.1})$$

where $u_{i,t}$ are independent white noise, satisfy $\sum_{k=1}^p \left(|\alpha_{i,k}| + \sum_{r=1}^{s_k} \sum_{c=1}^C |\beta_{kr,c}| \right) < 1$ for all $X_{i,t}$, and $i \in \mathcal{K}$. Then $\mathbf{X}_t$ is stationary.

Corollary 2. Let $\mathbf{X}_t$ be a community- α GNAR($[p_c], \{[s_k(c)]\}, [C], \mathcal{I}[C]$) process such that all of the coefficients in (8) satisfy

$$\sum_{k=1}^{p_c} \left\{ |\alpha_{k,c}| + \sum_{r=1}^{s_k(c)} \left(|\beta_{k,r,c}| + \sum_{\tilde{c} \in \mathcal{I}_c} |\gamma_{k,r,c;\tilde{c}}| \right) \right\} < 1, \quad (\text{C.2})$$

for all covariates $c \in [C]$. Then $\mathbf{X}_t$ is stationary.

See Knight et al. (2020) for a proof of Theorem 4 and the supplementary material Section I for a proof of Corollary 2. In view of Remark 1, we immediately derive a more general condition by exploiting the VAR representation in (10) and well-known conditions for stationarity of general VAR processes; see Chapter 2.1.1 in Lütkepohl (2005).

Corollary 3. Let $\mathbf{X}_t$ be a community- α GNAR $([p_c], \{[s_k(c)]\}, [C], \mathcal{I}[C])$ process such that the autoregressive matrices in (9) satisfy

$$\det \left\{ \mathbf{I}_d - \sum_{k=1}^p z^k \Phi_k(t-k) \right\} \neq 0, \quad (\text{C.3})$$

for all $|z| \leq 1$. Then, $\mathbf{X}_t$ given by the VAR representation in (10) is stationary.

We remark that condition (C.2) implies condition (C.3) and is easier to verify, and that these are sufficient but not necessary conditions, i.e., a GNAR process given by (8) could still be stationary if these are not satisfied.

D Optimisation algorithms and further examples

Algorithm 1 Simultaneous covariance and parameter estimation

Alternate until convergence:

Estimate the error covariance: $\hat{\Sigma}_{\mathbf{u}} \leftarrow (1/n) \sum_{t_i=1}^n (\mathbf{y}_{t_i} - \mathbf{R}_{t_i} \hat{\boldsymbol{\theta}})(\mathbf{y}_{t_i} - \mathbf{R}_{t_i} \hat{\boldsymbol{\theta}})^T$

Estimate the covariance of $\mathbf{u}$ in (13): $\hat{\Sigma}_T \leftarrow \mathbf{I}_n \otimes \hat{\Sigma}_{\mathbf{u}}$

Estimate $\boldsymbol{\theta}$ in (13) using generalised least squares: $\hat{\boldsymbol{\theta}} \leftarrow \left(\mathbf{R}^T \hat{\Sigma}_T^{-1} \mathbf{R} \right)^{-1} \mathbf{R}^T \hat{\Sigma}_T^{-1} \mathbf{y}$

Let `network_generator` denote a network estimation algorithm, e.g., sampling a network with d nodes from an Erdős-Rényi random graph model, and denote all loss functions as above.

Algorithm 2 Backnet-fitting with network estimation

Alternate until convergence or a threshold has been reached:

- 1: Generate a network from a fitted or prior-generative model

$$\hat{\mathcal{G}} \leftarrow \text{network_generator}(\mathbf{y}, \hat{\boldsymbol{\theta}}_Y)$$

- 2: Estimate the unknown parameters and coefficients in (18) using Algorithm 3 and the network-informed design $\mathbf{R}(\hat{\mathcal{G}})$

$$\hat{\boldsymbol{\theta}}_Y \leftarrow \text{backnet_fitting}\{\mathbf{y}, \mathbf{R}(\hat{\mathcal{G}})\}$$

- 3: Evaluate the one-step ahead forecast mean-squared error, store for comparison with other networks, and finish procedure if smaller than a predetermined threshold ϵ

$$\text{MSPE}(\hat{\boldsymbol{\theta}}_Y, \hat{\mathcal{G}}) \leftarrow \|\mathbf{y} - (\Psi_n \boldsymbol{\theta}_\mu + \Phi_n \boldsymbol{\theta}_\pi + \mathbf{R}(\hat{\mathcal{G}}) \boldsymbol{\theta}_X)\|_2^2$$

Let $\mathcal{L}_\mu$ and $\mathcal{L}_\pi$ denote our choice of penalised least squares loss functions to estimate the trend and seasonal component of each node, e.g., smoothing splines (Wahba 1990) or a Fourier basis expansion (Ramsay & Silverman 2010, Chapter 3).

Algorithm 3 Backnet-fitting

Alternate until convergence:

- 1: Use the partial residuals $\hat{\mathbf{u}}_\mu = \mathbf{y} - (\Phi_n \hat{\boldsymbol{\theta}}_\pi + \mathbf{Z} \hat{\boldsymbol{\eta}}_{[C]} + \mathbf{R}(\hat{\mathbf{x}}) \hat{\boldsymbol{\theta}}_X)$ to estimate $\boldsymbol{\theta}_\mu$

$$\hat{\boldsymbol{\theta}}_\mu \leftarrow \underset{\boldsymbol{\theta}_\mu}{\operatorname{argmin}} \{ \mathcal{L}_\mu(\hat{\mathbf{u}}_\mu; \boldsymbol{\lambda}_\mu) \}.$$

- 2: Use the partial residuals $\hat{\mathbf{u}}_\pi = \mathbf{y} - (\Psi_n \hat{\boldsymbol{\theta}}_\mu + \mathbf{Z} \hat{\boldsymbol{\eta}}_{[C]} + \mathbf{R}(\hat{\mathbf{x}}) \hat{\boldsymbol{\theta}}_X)$ to estimate $\boldsymbol{\theta}_\pi$

$$\hat{\boldsymbol{\theta}}_\pi \leftarrow \underset{\boldsymbol{\theta}_\pi}{\operatorname{argmin}} \{ \mathcal{L}_\pi(\hat{\mathbf{u}}_\pi; \boldsymbol{\lambda}_\pi) \}.$$

- 3: Use the partial residuals $\hat{\mathbf{x}} = \mathbf{y} - (\Psi_n \hat{\boldsymbol{\theta}}_\mu + \mathbf{Z} \hat{\boldsymbol{\eta}}_{[C]} + \Phi_n \hat{\boldsymbol{\theta}}_\pi)$ to estimate $\boldsymbol{\theta}_X$

$$\hat{\boldsymbol{\theta}}_X \leftarrow \underset{\boldsymbol{\theta}_X}{\operatorname{argmin}} \{ \mathcal{L}_X(\hat{\mathbf{x}}; \boldsymbol{\lambda}_X) \}.$$

- 4: Use the partial residuals $\hat{\mathbf{u}}_Y = \mathbf{y} - (\Psi_n \hat{\boldsymbol{\theta}}_\mu + \Phi_n \hat{\boldsymbol{\theta}}_\pi + \mathbf{R}(\hat{\mathbf{x}}) \hat{\boldsymbol{\theta}}_X)$ to estimate $\boldsymbol{\eta}_{[C]}$

$$\hat{\boldsymbol{\eta}}_{[C]} \leftarrow (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \hat{\mathbf{u}}_Y.$$

E Community structure estimation

There are two possible extremes, either each node is its own community, or the entire network is one community. The former corresponds to a local- $\alpha\beta$ GNAR model and the latter to a global- α one. Recall that a local- $\alpha\beta$ is given by (C.1), where each node has a different vector of coefficients $\boldsymbol{\theta}_i = (\alpha_{k,i}, \gamma_{k,r,i}, \dots, \gamma_{p,s_p,i})$. We express a community- α model in local- $\alpha\beta$ form by noting that if $i, j \in K_c$, then $\alpha_{k,i} = \alpha_{k,j}$ and $\gamma_{k,r,i} = \gamma_{k,r,j}$ at all lags k and r -stages, i.e., communities *share* the same parameter values. Thus, we can fuse local parameter estimates to obtain community membership estimates.

Consider the *fused group lasso* criterion

$$\hat{\boldsymbol{\theta}}_{\text{fgl}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \left\{ \left\| \mathbf{y} - \sum_{i=1}^d \mathbf{R}_i \boldsymbol{\theta}_i \right\|_2^2 + \lambda_C \sum_{i < j} \|\boldsymbol{\theta}_i - \boldsymbol{\theta}_j\|_2 \right\}, \quad (\text{E.1})$$

where we fix the same model order for all nodes, i.e., $\boldsymbol{\theta}_i$ is q -dimensional for all $i \in [d]$, $\mathbf{R}_i$ is the i th 'design' matrix, $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_d)$ is the vector of unknown parameters, and $\|\boldsymbol{\theta}_i - \boldsymbol{\theta}_j\|_2$ is the ℓ_2 -norm of differences between the estimated coefficients. The solutions, $\hat{\boldsymbol{\theta}}_{\text{fgl}}$, to (E.1) depend on the hyperparameter $\lambda_C \geq 0$ and have attractive properties. Suppose we are estimating the community-structure of a ten-dimensional network time series, then a solution to (E.1) with three communities would be

$$\hat{\boldsymbol{\theta}}_{\text{fgl}} = \left(\underbrace{\hat{\boldsymbol{\theta}}_1, \hat{\boldsymbol{\theta}}_1}_{\hat{K}_1}, \underbrace{\hat{\boldsymbol{\theta}}_2, \hat{\boldsymbol{\theta}}_2, \hat{\boldsymbol{\theta}}_2}_{\hat{K}_2}, \underbrace{\hat{\boldsymbol{\theta}}_3, \hat{\boldsymbol{\theta}}_3, \hat{\boldsymbol{\theta}}_3, \hat{\boldsymbol{\theta}}_3, \hat{\boldsymbol{\theta}}_3}_{\hat{K}_3} \right),$$

where $\min\{\|\hat{\boldsymbol{\theta}}_c - \hat{\boldsymbol{\theta}}_{\tilde{c}}\|_2\} > \lambda_C$ for $c, \tilde{c} \in [\hat{C} = 3]$. Thus, communities share parameter values.

We can think of the penalty in (E.1) as a constraint on the ℓ_1 -norm of the ℓ_2 -norm differences between estimates for θ_i and θ_j , i.e., $\sum_{i < j} \|\Delta_{ij}\|_2 \leq b_C$, where $\Delta_{ij} := \theta_i - \theta_j$ and $b_C \geq 0$ is the ℓ_1 -norm constraint linked to the Lagrange multiplier λ_C . Fix all indices except i and j , and recall that the subgradient of the ℓ_2 -norm, ω_{ij} , is $\Delta_{ij}/\|\Delta_{ij}\|_2$ if $\Delta_{ij} \neq \mathbf{0}$ and if $\Delta_{ij} = \mathbf{0}$, then ω_{ij} is any q -dimensional vector such that $\|\omega_{ij}\|_2 \leq 1$ (Hastie et al. 2015, Chapter 4.1). If $\widehat{\Delta}_{ij} \neq \mathbf{0}$, then solutions to (E.1) with the simpler constraint of fusing sequential indices satisfy the subgradient equations

$$\begin{aligned}\hat{\theta}_i &= \left(\mathbf{R}_i^T \mathbf{R}_i + \frac{\lambda_C}{\|\widehat{\Delta}_{ij}\|_2} \mathbf{I}_q \right)^{-1} \left(\mathbf{R}_i^T \hat{\mathbf{u}}_i + \frac{\lambda_C}{\|\widehat{\Delta}_{ij}\|_2} \hat{\theta}_j \right), \\ \hat{\theta}_j &= \left(\mathbf{R}_j^T \mathbf{R}_j + \frac{\lambda_C}{\|\widehat{\Delta}_{ij}\|_2} \mathbf{I}_q \right)^{-1} \left(\mathbf{R}_j^T \hat{\mathbf{u}}_j + \frac{\lambda_C}{\|\widehat{\Delta}_{ij}\|_2} \hat{\theta}_i \right),\end{aligned}\tag{E.2}$$

where $\hat{\mathbf{u}}_i = \mathbf{y} - \sum_{\ell \neq i} \mathbf{R}_\ell \hat{\theta}_\ell$ are the i th partial residuals of a local- $\alpha\beta$ model. If $\|\widehat{\Delta}_{ij}\|_2$ is large, then both equations in (E.2) are approximately equal to a local least squares estimator, i.e., $\hat{\theta}_i \approx (\mathbf{R}_i^T \mathbf{R}_i)^{-1} \mathbf{R}_i^T \hat{\mathbf{u}}_i$. Whereas, if $\|\widehat{\Delta}_{ij}\|_2$ is small, then the identity term dominates, and we have that $\hat{\theta}_i \approx \hat{\theta}_j$. Now, if $\widehat{\Delta}_{ij} = \mathbf{0}$, then a subgradient of (E.1) can be written as

$$(\mathbf{R}_i^T \mathbf{R}_i)^{-1} \mathbf{R}_i^T \hat{\mathbf{u}}_i - (\mathbf{R}_j^T \mathbf{R}_j)^{-1} \mathbf{R}_j^T \hat{\mathbf{u}}_j + \widehat{\Delta}_{ij} = \lambda_C \omega_{ij},\tag{E.3}$$

where $\|\omega_{ij}\|_2 \leq 1$. Hence, we have a lasso-like property for the local least squares estimators. That is, if $\widehat{\Delta}_{ij} = \mathbf{0}$, then by (E.3) we have that

$$\|(\mathbf{R}_i^T \mathbf{R}_i)^{-1} \mathbf{R}_i^T \hat{\mathbf{u}}_i - (\mathbf{R}_j^T \mathbf{R}_j)^{-1} \mathbf{R}_j^T \hat{\mathbf{u}}_j\|_2 \leq \lambda_C.\tag{E.4}$$

Notice that if $\lambda_C = 0$, then (E.1) is an unconstrained problem and each node is its own community (local- $\alpha\beta$), no matter how small $\|\widehat{\Delta}_{ij}\|_2$ is, whereas, if $\lambda_C = +\infty$, then all nodes belong to the same community (global- α), regardless of how large $\|\widehat{\Delta}_{ij}\|_2$ is. Hence, our choice of λ_C automatically determines the number of clusters. Moreover, the system in (E.2) can be solved by iterating a gradient descent algorithm, and the full subgradient is given by the sum of all node pairs, i.e., adding the right hand side in (E.2) for all $j \neq i$ in the first equation and for all $i \neq j$ in the second one. Due to space constraints and computational efficiency we consider an alternative dynamic programming algorithm that produces solutions to (E.1).

By construction we have that the partial residuals in (E.4) satisfy $\hat{\mathbf{u}}_i = \mathbf{y}_i$, where $\mathbf{y}_i = (X_{i,p+1}, \dots, X_{i,T})$. Let $\tilde{\theta}_i$ denote the i th least squares estimator in a local- $\alpha\beta$ model, i.e., (14) for each $i \in [d]$. Then, the constraint in (E.4) can be written as $\|\tilde{\theta}_i - \tilde{\theta}_j\|_2 \leq \lambda_C$. Now, there are d local- $\alpha\beta$ least squares estimators and by Theorem 1 and Corollary 1 these are also consistent estimators of community-wise coefficients. Thus, we can think of these d values as a finite collection of possible solutions to (E.1), which motivates formalising this as a constraint, i.e., $\hat{\theta}_{\text{fgl}} = (\hat{\theta}_1, \dots, \hat{\theta}_d)$ given by (E.1) such that each $\hat{\theta}_i \in \{\tilde{\theta}_i\}$, where $i = 1, \dots, d$. Algorithm 4 solves (E.1) with this additional constraint.

Algorithm 4 Dynamic programming for the fused grouped lasso

Initialise: Number of communities: $\widehat{C} \leftarrow 0$, set of nonassigned nodes: $A \leftarrow [d]$.

Estimate $\boldsymbol{\theta}_i$ for each node in a local- $\alpha\beta$ design:

$$\tilde{\boldsymbol{\theta}}_i \leftarrow (\mathbf{R}_i^T \mathbf{R}_i)^{-1} \mathbf{R}_i^T \mathbf{y}_i.$$

while $A \neq \emptyset$ **do:**

 Select the smallest index that has not been assigned: $i \leftarrow \min_i \{i : i \in A\}$.

 Increase the number of communities by one: $\widehat{C} \leftarrow \widehat{C} + 1$.

 Initialise the community of node i : $\widehat{K}_{\widehat{C}} \leftarrow \{i\}$.

 Update the set of nonassigned nodes: $A \leftarrow A \setminus \{i\}$.

 Select all nodes that belong to $\widehat{K}_{\widehat{C}}$. That is, for $j \in A$ and $j \neq i$

if $\|\tilde{\Delta}_{ij}\|_2 \leq \lambda_C$, **then:** ▷ Update communities

$$\widehat{K}_{\widehat{C}} \leftarrow \widehat{K}_{\widehat{C}} \cup \{j\},$$

$$A \leftarrow A \setminus \{j\}.$$

Return: $\{\widehat{K}_c, \hat{\boldsymbol{\theta}}_c\}$, for $c \in [\widehat{C}]$. ▷ Community structure

E.1 Including interactions into the fused grouped lasso

Let $\widehat{K}_c$ denote estimated communities, and introduce interaction terms into (E.1),

$$\hat{\boldsymbol{\theta}}_{\text{fgl}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \left\{ \|\mathbf{y} - \sum_{c=1}^{\widehat{C}} \mathbf{R}_c \boldsymbol{\theta}_c\|_2^2 + \lambda_C \sum_{c < \tilde{c}} \|\boldsymbol{\theta}_c - \boldsymbol{\theta}_{\tilde{c}}\|_2 + \lambda_I \sum_{c < \tilde{c}} \|\boldsymbol{\theta}_{c:\tilde{c}}\|_1 \right\}, \quad (\text{E.5})$$

where $\boldsymbol{\theta}_{c:\tilde{c}} = (\gamma_{k,r,c:\tilde{c}})$ for $k = 1, \dots, p$ and $r = 1, \dots, s_p$ are the unknown *interaction* coefficients in (7). Solutions to (E.5) cluster nodes according to parameter values and produce sparse interactions coefficients, i.e., $\hat{\boldsymbol{\theta}}_{c:\tilde{c}} = \mathbf{0}$. Due to space constraints, we do not pursue this method further.

F Algorithms for fitting regularised estimators

$\mathcal{L}_c(\mathbf{y}_c, \mathbf{R}_c; \lambda) = \sum_{i=1}^{n_c} \mathcal{L}_c(y_{i,c}, [\mathbf{R}_c]_i; \lambda)$, where

$$\mathcal{L}_c(y_{i,c}, [\mathbf{R}_c]_i; \lambda_c) = \begin{cases} (y_{i,c} - \langle [\mathbf{R}_c]_i, \boldsymbol{\theta}_c \rangle)^2, & \text{if } |y_{i,c} - \langle [\mathbf{R}_c]_i, \boldsymbol{\theta}_c \rangle| \leq \lambda_c \\ \lambda_c (2|y_{i,c} - \langle [\mathbf{R}_c]_i, \boldsymbol{\theta}_c \rangle| - \lambda_c), & \text{otherwise} \end{cases}, \quad (\text{F.1})$$

$\lambda_c > 0$ is the Huber loss “smoothing” hyperparameter and reflects our outlier “tolerance”, $[\mathbf{R}_c]_i$ is the i th row of $\mathbf{R}_c$ and $y_{i,c}$ is the i th “response” value for community K_c ; see XX in the supplementary material for an alternating algorithm that minimises (F.1).

Algorithm 5 Simultaneous covariance and parameter estimation for high-dimensional data

Alternate until convergence:

Estimate the error covariance: $\hat{\Sigma}_{\mathbf{u}} \leftarrow \underset{\Sigma_{\mathbf{u}}}{\operatorname{argmin}} \{ \mathcal{L}_{\Sigma}(\hat{\mathbf{u}}) \}.$

Estimate the covariance of $\mathbf{u}$ in (13): $\hat{\Sigma}_T \leftarrow \mathbf{I}_n \otimes \hat{\Sigma}_{\mathbf{u}}.$

Estimate $\boldsymbol{\theta}$ accounting for correlated errors: $\hat{\boldsymbol{\theta}} \leftarrow \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \left\{ \mathcal{L}_{\boldsymbol{\theta}}(\hat{\Sigma}_T^{-1/2} \mathbf{y}, \hat{\Sigma}_T^{-1/2} \mathbf{R}; \boldsymbol{\lambda}) \right\}.$

G Simulating multiple GNAR processes

This section explores our methods via multiple simulation studies. For illustration purposes we employ the USA state-wise network; see Figure 2, as the underlying structure, and analyse multiple realisations from different network time series models. Subsequently, we use the model selection and diagnostic tools suggested by Nason et al. (2025) for selecting model order for a given network time-series and we use Akaike’s information criteria (AIC) to assess goodness of fit.

We simulate one thousand instances of different community- α GNAR processes using the USA state-wise network with time-varying weights of period four (i.e., each weight function is seasonal with a four year period). We illustrate Theorem 1 by looking at the ℓ_2 -error-norm of the estimator (14) and a true parameter vector, which we change at every iteration. All simulated GNAR processes in this subsection are given by

$$\mathbf{X}_t = \boldsymbol{\alpha}(\mathbf{X}_t; [3, 2, 1], [3]) + \boldsymbol{\beta}\{\mathbf{X}_t; \{[2, 2, 1], [3, 2], [3]\}, [3]\} + \cdots \\ \gamma\{\mathbf{X}_t; \{(3), (3), (1, 2)\}, [3]\} + \mathbf{u}_t, \quad (\text{G.1})$$

where $\mathbf{u}_t \sim N_d(0, \mathbf{I}_d)$ are independent at all times t , and $d = 51$. Model (G.1) is a community- α GNAR with interactions for three communities and fifty-one nodes. The model order for community one is $(3, [2, 2, 1], \{3\})$ and interactions with community three, i.e., $\gamma_{\cdot, 1:2} \equiv 0$, for community two it is $(2, [3, 2], \{3\})$ and interactions with community three, i.e., $\gamma_{\cdot, 2:1} \equiv 0$, and for community three it is $(1, [3], \{1, 2\})$ and interactions with communities one and three (all possible r -stage neighbours). The time-varying weights are given by

$$\begin{aligned} w_{ij}(t) &= \{1 + \cos(t\pi/2) + 0.1\} 2^{-d_{ij}/2} \mathbb{I}(j \in K_1), \\ w_{ij}(t) &= \{1 + \sin(t\pi/2) + 0.1\} 2^{-d_{ij}} \mathbb{I}(j \in K_2), \\ w_{ij}(t) &= \{1 + \cos(t\pi/4) \sin(t\pi/4) + 0.1\} 2^{-d_{ij}} \mathbb{I}(j \in K_3), \end{aligned} \quad (\text{G.2})$$

where $d_{ij} \in \{1, \dots, 11\}$ is the shortest finite path length between nodes i and j in the USA state-wise network. Each simulated realisation is produced from a valid $\boldsymbol{\theta}_0$, chosen to ensure a stationary simulated process. The experiment is summarised in psuedo-code in Algorithm 6 below.

Algorithm 6 Simulation study

```

1: set.seed(2024)
2: for  $T \in \{1, \dots, 1000\}$  do
3:    $\theta_0 \leftarrow \text{simulate}(\theta)$   $\triangleright$  Simulate a stationary  $\theta_0$ , model order given by (G.1)
4:    $\mathbf{W}_t \leftarrow w_{ij}(t)$  for  $t = 1, \dots, T$   $\triangleright$  weight functions are given by (G.2)
5:    $\{\mathbf{X}_t\} \leftarrow \text{community\_GNARsim}(\theta_0, T)$   $\triangleright$  Simulate  $T$  time-steps of (G.1), with  $\theta_0$ 
6:    $\hat{\theta} \leftarrow (\mathbf{R}^T \mathbf{R})^{-1} \mathbf{R}^T \mathbf{y}$   $\triangleright$  Compute least-squares estimator given by (14)
7:   for  $c = 1, 2, 3$  do
8:      $\delta_c^{(T)} \leftarrow \|\hat{\theta}_c - \theta_{0,c}\|_2$   $\triangleright$  Compute community  $\ell_2$ -error-norm
9:   end for
10:   $\delta^{(T)} \leftarrow \|\hat{\theta} - \theta_0\|_2$   $\triangleright$  Save  $\ell_2$ -error-norm for the whole model
11: end for
12: return:  $\{(\delta_1^{(T)}, \delta_2^{(T)}, \delta_3^{(T)}, \delta^{(T)})\}$ 

```

Figure 4 shows the results. Code for replicating the study can be found in the CRAN GNAR package in due course.

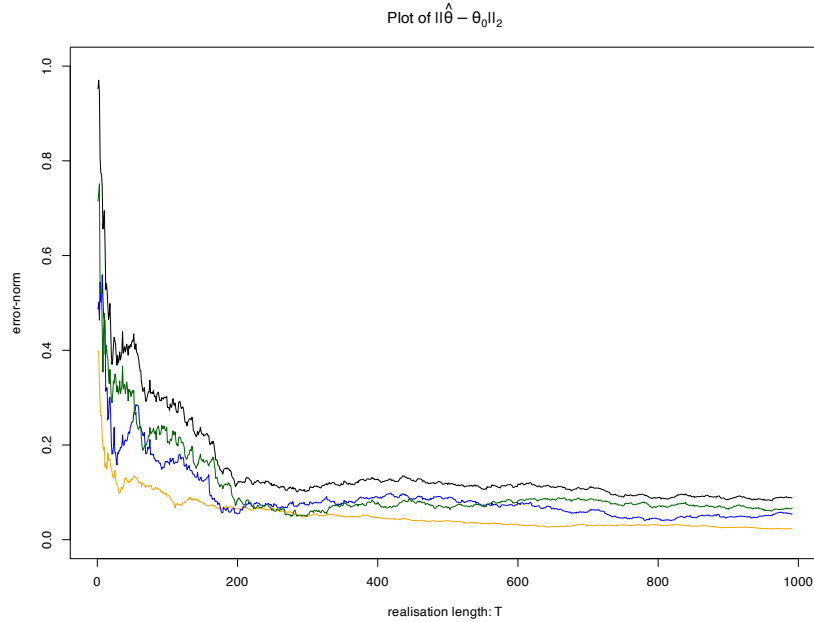


Figure 4: Multiple simulations of stationary GNAR processes given by (G.1). Each simulated realisation has a different length $T = 1, \dots, 1000$, and “true” parameter vector θ_0 ; see Algorithm 6. Each curve is ℓ_2 -error-norm, **orange** is for K_1 , set of nodes identified as *Red states*, **blue** is for K_2 , (*Blue states*), **green** is for K_3 , (*Swing states*), and black is for the whole model.

We close this subsection by noting that all four error curves in Figure 4 decrease jointly towards zero as realisation length T increases. This illustrates the results in Theorem 1, and shows the reasonable performance of $\hat{\theta}$ for realisations shorter than one-hundred time-steps with thirty-five unknown parameters. Table 4 shows the estimated coefficients for our chosen model GNAR $([1, 2], \{[1], [1, 1]\}, 2)$.

Table 4: Mean estimator values for only 10 simulations, each of length 100 coming from a GNAR $([1, 2], \{[1], [1, 1]\}, 2)$. **Mean Est.** is the mean estimator value, **Sim. Sd.** is the standard deviation of the 10 simulations (standard error), and **True Value** is the known parameter. The simulations are computed using random seeds 1983 to 1993 in R.

	$\hat{\alpha}_{11}$	$\hat{\beta}_{111}$	$\hat{\alpha}_{12}$	$\hat{\beta}_{112}$	$\hat{\alpha}_{22}$	$\hat{\beta}_{212}$
Mean Est.	0.280	0.177	0.222	0.302	0.115	0.180
Sim. Sd.	0.062	0.109	0.061	0.098	0.083	0.127
True Value	0.270	0.180	0.250	0.300	0.120	0.200

Table 4 shows the reasonable performance of our estimator $\hat{\theta}$.

H Analysis of one-lag differenced data

The estimated coefficients in Table 2 do not satisfy the stationarity conditions in Corollary 2, however, these are sufficient, but not necessary.

Hence, it is prudent to additionally study the one-lag differenced series. The R-Corbit plot in Figure 5 suggests that one-lag differencing removes correlation across all r -stages for the first lag, and that the PNACF seems to cut off after lag three, and that it does not cut-off at particular r -stages. We hence fit a community- α GNAR(3, $\{[1, 1, 1]\}, 3$) to the one-lag differenced data with estimates shown in Table 5.

Table 5: Approx. statistically significant coefficients at the 5% level estimated with the standardised one-lag differenced network time series $(\tilde{\mathbf{X}}_t, \mathcal{G})$ of vote percentages for the Republican nominee in presidential elections in the USA from 1980 to 2020. The fit is a community- α GNAR(3, $\{[1, 1, 1]\}, 3$). [Recall from (8) that the coefficients are α_{kc} and β_{krc} , where k is lag, r is network r -stage and c is community.]

	$\hat{\alpha}_{11}$	$\hat{\beta}_{111}$	$\hat{\alpha}_{21}$	$\hat{\alpha}_{31}$	$\hat{\alpha}_{12}$	$\hat{\beta}_{112}$	$\hat{\alpha}_{22}$	$\hat{\alpha}_{32}$	$\hat{\alpha}_{23}$	$\hat{\alpha}_{33}$
Estimate	-0.27	0.59	-0.48	-0.37	-0.40	0.57	-0.57	-0.36	-0.45	-0.24
Std. Err.	0.12	0.25	0.10	0.11	0.13	0.23	0.11	0.12	0.09	0.11

Interestingly, *Blue* and *Red* network effects remain after one-lag differencing, whereas, *Swing* state network effects are not approx statistically significant at the 5% level. This suggests that state-wise communities might have different second-order effects.

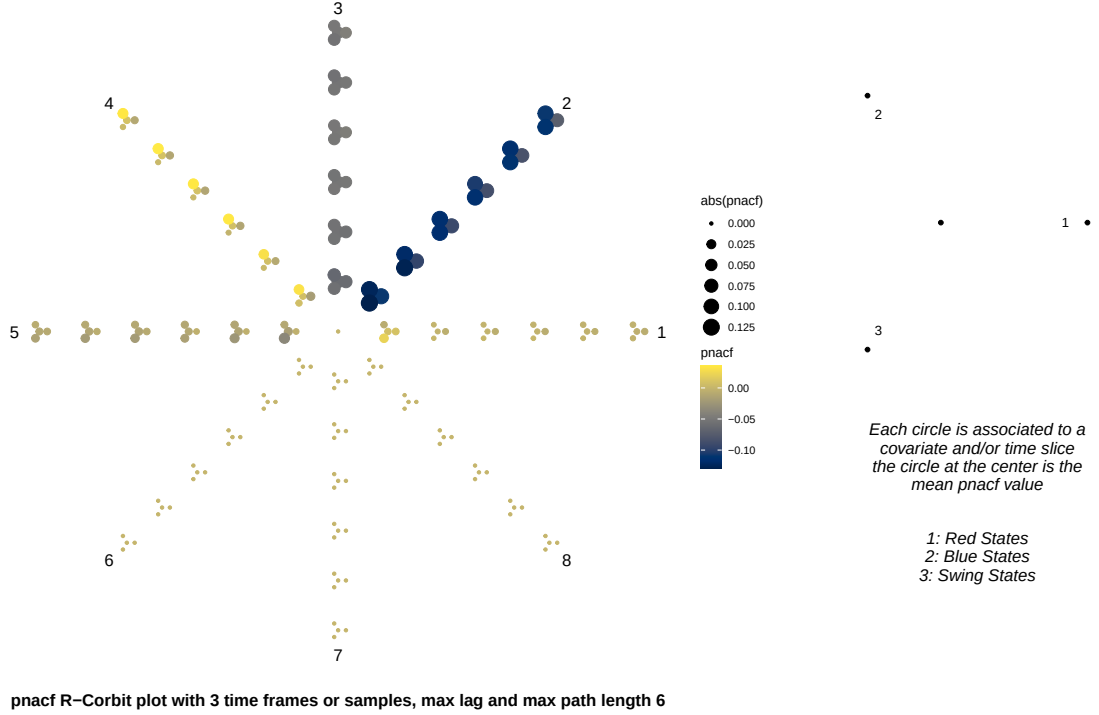


Figure 5: R-Corbit plot for the one-lag differenced network time series $\mathbf{X}_t - \mathbf{X}_{t-1}$. Each $X_{i,t}$ is the difference in the percentage of votes for the Republican nominee in the current election minus the previous one from 1980 to 2020. The communities are *Red*, *Blue* and *Swing* states; see Figure 2.

I Stationary community- α GNAR models

This is an immediate consequence of stationarity conditions for local- α GNAR models. We proceed to show Corollary 2.

Proof. We express the sum of stationary community processes as a local- α GNAR process as follows.

1. Let $\phi_{i,k} = \alpha_{k,c} \mathbb{I}(i \in K_c)$, where we set $\alpha_{k,c} = 0$ if $k > p_c$,
2. $\zeta_{i,k,r} = \beta_{k,r,c} \mathbb{I}(i \in K_c)$, where we set $\beta_{k,r,c} = 0$ if $k > p_c$ or $s_k(c) > r^*$,
3. $\eta_{i,k,c;\tilde{c}} = \gamma_{k,r,c;\tilde{c}} \mathbb{I}(i \in K_c)$, where we set $\gamma_{k,r,c;\tilde{c}} = 0$ if $k > p_c$, or $s_k(c) > r_c^*$, or $\tilde{c} \notin \mathcal{I}_c$.

Hence, the community with the largest lag will have the biggest number of non-zero autoregressive coefficients, and might have the most non-zero r -stage neighbourhood coefficients. Set $p = \max_{c \in [C]}([p_c])$ and $s_k = \max_{c \in [C]}([s_k(c)])$ for each $k = 1, \dots, p$. Then, the node-wise representation of the community- α GNAR model is given by

$$X_{i,t} = \sum_{k=1}^p \left\{ \phi_{i,k} X_{i,t-k} + \sum_{r=1}^{s_k} \left(\zeta_{i,k,r} Z_{i,t-k}^{r,c} + \sum_{\tilde{c} \in \mathcal{I}_c} \eta_{i,k,r} Z_{i,t-k}^{r,c;\tilde{c}} \right) \right\} + u_{i,t}, \quad (\text{I.1})$$

where $u_{i,t}$ are independent white noise and $Z_{i,t-k}^{r,c}$, $Z_{i,t-k}^{r,c;\tilde{c}}$ are the c -community r -stage neighbourhood regressions. For all lags $k \in \{1, \dots, p\}$ we have that for each $r \in \{1, \dots, s_k\}$ the node-wise within community r -stage neighbourhood regression coefficients for $X_{i,t}$ satisfy

$$\sum_{r=1}^{s_k} |\zeta_{i,k,r}| = \sum_{r=1}^{s_k} |\beta_{k,r,c}| \mathbb{I}(i \in K_c),$$

and that the between community r -stage neighbourhood regression coefficients for $X_{i,t}$ satisfy

$$\sum_{r=1}^{s_k} \sum_{\tilde{c} \in \mathcal{I}_c} |\eta_{i,k,r}| = \sum_{r=1}^{s_k} \sum_{\tilde{c} \in \mathcal{I}_c} |\gamma_{k,r,c;\tilde{c}}| \mathbb{I}(i \in K_c),$$

and that the autoregressive coefficients satisfy

$$\sum_{k=1}^p |\phi_{i,k}| = \sum_{k=1}^p |\alpha_{k,c}| \mathbb{I}(i \in K_c).$$

By construction we have that for all $X_{i,t}$ such that $i \in K_c$ the within community r -stage neighbourhood coefficients are equal for all lags and at all r -stages, i.e., if $i \in K_c$, then $\zeta_{i,k,r} = \beta_{k,r,c}$ otherwise $\zeta_{i,k,r} = 0$. Note that the same holds for between community coefficients, i.e., if $i \in K_c$, then $\eta_{i,k,r} = \gamma_{k,r,c;\tilde{c}}$ otherwise $\eta_{i,k,r} = 0$. Thus, the coefficients for each nodal time series $X_{i,t}$ satisfy

$$\begin{aligned} & \sum_{k=1}^p \left\{ |\phi_{i,k}| + \sum_{r=1}^{s_k} \sum_{c=1}^C (|\zeta_{i,k,r}| + |\eta_{i,k,r}|) \right\} \\ &= \mathbb{I}(i \in K_c) \sum_{k=1}^p \left\{ |\alpha_{k,c}| + \sum_{r=1}^{s_k} \left(|\beta_{k,r,c}| + \sum_{\tilde{c} \in \mathcal{I}_c} |\gamma_{k,r,c;\tilde{c}}| \right) \right\}. \end{aligned} \quad (\text{I.2})$$

Assume that

$$\sum_{k=1}^{p_c} \left\{ |\alpha_{k,c}| + \sum_{r=1}^{s_k(c)} \left(|\beta_{k,r,c}| + \sum_{\tilde{c} \in \mathcal{I}_c} |\gamma_{k,r,c;\tilde{c}}| \right) \right\} < 1, \quad (\text{I.3})$$

holds for all covariates $c \in [C]$. Combining this with (I.1) and (I.2) we have that each local- α (i.e., $\phi_{i,k}$) nodal process satisfies

$$\sum_{k=1}^p \left\{ |\phi_{i,k}| + \sum_{r=1}^{s_k} \sum_{c=1}^C (|\zeta_{i,k,r}| + |\eta_{i,k,r}|) \right\} < 1,$$

for all $i \in \mathcal{K}$. Therefore, by Theorem 4 the process given by (I.1) is stationary. Recalling that (I.1) is identical to the process given by (8) finishes the proof. $\blacksquare$

J Nonasymptotic bounds

We present a proof of Theorem 1. We begin by recalling the fixed design we obtain by conditioning the observed realisation, recall from Section 3 that a realisation of length

$T \in \mathbb{Z}^+$ coming from a stationary community- α GNAR process can be expressed as the linear model (13), i.e.,

$$\mathbf{y} = \mathbf{R}\boldsymbol{\theta}_0 + \mathbf{u},$$

where $\mathbf{u} = \sum_{c=1}^C \mathbf{u}_c$ is a vector of independent zero-mean random variables, $\mathbf{y} \in \mathbb{R}^{d(T-p)}$ is the response, $\mathbf{R} \in \mathbb{R}^{d(T-p) \times q}$ is the design matrix, $\boldsymbol{\theta}_0 \in \mathbb{R}^q$, where $q = \sum_{c=1}^C q_c$, and that we estimate the “true” vector of parameters $\boldsymbol{\theta}_0 = (\boldsymbol{\theta}_{0,1}, \dots, \boldsymbol{\theta}_{0,C})$ by ordinary least-squares.

J.1 Lower bound for the community norm

We begin by inspecting A.1, recall that

$$\mathbf{R} = [\mathbf{R}_1 | \dots | \mathbf{R}_C],$$

and notice that each c -community design matrix has zeros in different row entries in $\mathbf{R}_c$, hence, without loss of generality we can write

$$\mathbf{R}^T \mathbf{R} = \begin{pmatrix} \mathbf{R}_1^T \mathbf{R}_1 & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \mathbf{R}_C^T \mathbf{R}_C \end{pmatrix},$$

i.e., $\mathbf{R}^T \mathbf{R}$ is diagonal by blocks.

Thus, if $\text{rank}(\mathbf{R}_c) = q_c$ for all $c \in [C]$ (i.e., each community linear model is not underdetermined), then a singular value decomposition of $\mathbf{R} = \mathbf{U} \mathbf{D} \mathbf{V}^T$, where $\mathbf{U} \in \mathbb{R}^{n \times n}$ is an orthonormal matrix, $\mathbf{D} \in \mathbb{R}^{n \times q}$ is a rectangular diagonal matrix with $q > 0$ non-zero entries (positive singular values) in the diagonal, and $\mathbf{V} \in \mathbb{R}^{q \times q}$ is an orthonormal matrix, can be expressed by blocks, i.e., $\mathbf{R}_c = \mathbf{U}_c \mathbf{D}_c \mathbf{V}_c^T$, and satisfies for all $\boldsymbol{\theta} \in \mathbb{R}^q$ the following

$$\begin{aligned} \|\mathbf{R}\boldsymbol{\theta}\|_2^2 &= \boldsymbol{\theta}^T (\mathbf{R}^T \mathbf{R}) \boldsymbol{\theta} \\ &= \sum_{c=1}^C \boldsymbol{\theta}_c^T (\mathbf{R}_c^T \mathbf{R}_c) \boldsymbol{\theta}_c \\ &= \sum_{c=1}^C \|\mathbf{D}_c \mathbf{V}_c^T \boldsymbol{\theta}_c\|_2^2 \end{aligned}$$

where $\mathbf{D}_c$ and $\mathbf{V}_c^T$ are the block matrices corresponding to community K_c .

• By applying assumption A.1 we have that each non-zero entry in $\mathbf{D}_c$ is lower bounded by $\{\tau_c |K_c| (T - p_c)\}^{1/2}$. Thus, we have that

$$\|\mathbf{D}_c \mathbf{V}_c^T \boldsymbol{\theta}_c\|_2^2 \geq \{\tau_c |K_c| (T - p_c)\}^{1/2} \|\mathbf{V}_c^T \boldsymbol{\theta}_c\|_2^2,$$

for all $c \in [C]$.

• Since $\mathbf{V}_c^T$ is an orthonormal matrix, i.e., $\|\mathbf{V}_c^T \boldsymbol{\theta}_c\|_2 = \|\boldsymbol{\theta}_c\|_2$, we have that

$$\|\mathbf{D}_c \mathbf{V}_c^T \boldsymbol{\theta}_c\|_2^2 \geq \{\tau_c |K_c| (T - p_c)\}^{1/2} \|\boldsymbol{\theta}_c\|_2^2,$$

for all $c \in [C]$.

• Define $\tau := \min_{c \in [C]} (\tau_c)$, and $\mathcal{K}_{[C]} := \min_{c \in [C]} \{|K_c|\}$, i.e., the size of the smallest community, by the above, for all $c \in [C]$, we get the following lower bound

$$\|\mathbf{R}_c \boldsymbol{\theta}_c\|_2^2 = \|\mathbf{D}_c \mathbf{V}_c^T \boldsymbol{\theta}_c\|_2^2 \geq \{\tau \mathcal{K}_{[C]} (T - p)\}^{1/2} \|\boldsymbol{\theta}_c\|_2^2.$$

Thus, for all $\boldsymbol{\theta}_c \neq \mathbf{0}$, we have that

$$\{\tau\mathcal{K}_{[C]}(T-p)\}^{1/2}\|\boldsymbol{\theta}_c\|_2^2 \leq \{\tau_c|K_c|(T-p_c)\}^{1/2}\|\boldsymbol{\theta}_c\|_2^2 \leq \|\mathbf{R}_c\boldsymbol{\theta}_c\|_2^2. \quad (\text{J.1})$$

J.2 Upper bounding the community error vector norm

Now recall that the estimator given by (14) is the unique solution of

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\{\|\mathbf{y} - \mathbf{R}\boldsymbol{\theta}\|_2^2\}.$$

Hence, we have that

$$\|\mathbf{y} - \mathbf{R}\hat{\boldsymbol{\theta}}\|_2^2 \leq \|\mathbf{y} - \mathbf{R}\boldsymbol{\theta}_0\|_2^2,$$

which after expanding the left-hand side, recalling that $\mathbf{u} = \mathbf{y} - \mathbf{R}\boldsymbol{\theta}_0$ and then rearranging gives the *basic inequality*

$$\|\mathbf{R}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)\|_2^2 \leq 2\langle \mathbf{u}, \mathbf{R}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \rangle.$$

Hence, by noting that each $\hat{\boldsymbol{\theta}}_c$ corresponds to different entries in $\mathbf{y}$ (i.e., $\boldsymbol{\xi}_c \odot \hat{\boldsymbol{\theta}}_c$ are orthogonal), we have that

$$\sum_{c=1}^C \|\mathbf{R}_c(\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c})\|_2^2 \leq 2 \sum_{c=1}^C \langle \mathbf{u}, \mathbf{R}_c(\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c}) \rangle, \quad (\text{J.2})$$

where $\boldsymbol{\theta}_{0,c}$ are the values of $\boldsymbol{\theta}_0$ corresponding to community c .

By equations (J.1) and (J.2) for each community K_c , we have that

$$\begin{aligned} \|\mathbf{R}_c(\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c})\|_2^2 &\leq 2\langle \mathbf{u}, \mathbf{R}_c(\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c}) \rangle \\ &= 2\langle \mathbf{R}_c^T \mathbf{u}, (\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c}) \rangle \\ &\stackrel{(i)}{\leq} 2\|\mathbf{R}_c^T \mathbf{u}\|_\infty \|(\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c})\|_1 \\ &\stackrel{(ii)}{\leq} 2 \left\{ p_c + \sum_{k=1}^{p_c} s_k(c)(1 + |\mathcal{I}_c|) \right\}^{1/2} \|\mathbf{R}_c^T \mathbf{u}\|_\infty \|(\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c})\|_2, \end{aligned} \quad (\text{J.3})$$

above (i) follows from Holder's inequality using the ℓ_1 - and ℓ_∞ -norms as conjugates, and (ii) follows from $\ell_1 - \ell_2$ -norm inequality in $\mathbb{R}^{q_c}$. By applying (J.1) to (J.3) with respect to the error vector $(\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c})$ we have that

$$\{\tau\mathcal{K}_{[C]}(T-p)\}^{1/2}\|(\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c})\|_2^2 \leq 2 \left\{ p_c + \sum_{k=1}^{p_c} s_k(c)(1 + |\mathcal{I}_c|) \right\}^{1/2} \|\mathbf{R}_c^T \mathbf{u}\|_\infty \|(\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c})\|_2,$$

or

$$\|(\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c})\|_2 \leq 2 \left\{ p_c + \sum_{k=1}^{p_c} s_k(c)(1 + |\mathcal{I}_c|) \right\}^{1/2} \{\tau\mathcal{K}_{[C]}(T-p)\}^{-1/2} \|\mathbf{R}_c^T \mathbf{u}\|_\infty. \quad (\text{J.4})$$

J.3 Jointly bounding the error for all communities

We are ready to prove the first part of Theorem 1.

Proof. Before bounding the error we note that each community-wise vector norm is the same if we augment the vector with zeros in the entries that correspond to members that do not belong to community K_c , i.e., $\|\boldsymbol{\theta}_c\|_2^2 = \|\boldsymbol{\xi}_c \odot \boldsymbol{\theta}\|_2^2$, hence, by orthogonality of the $\boldsymbol{\xi}_c$, we have that $\|\boldsymbol{\theta}\|_2^2 = \sum_{c=1}^C \|\boldsymbol{\theta}_c\|_2^2$. The community-wise decomposition allows jointly bounding the error vector for all communities as follows.

$$\begin{aligned} \|(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)\|_2^2 &= \sum_{c=1}^C \|(\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}_{0,c})\|_2^2 \\ &\stackrel{(i)}{\leq} \sum_{c=1}^C \left[2 \left\{ p_c + \sum_{k=1}^{p_c} s_k(c)(1 + |\mathcal{I}_c|) \right\}^{1/2} \{\tau \mathcal{K}_{[C]}(T - p)\}^{-1/2} \|\mathbf{R}_c^T \mathbf{u}\|_\infty \right]^2 \\ &\stackrel{(ii)}{\leq} [2(Cq_{[C]})^{1/2} \{\tau \mathcal{K}_{[C]}(T - p)\}^{-1/2} \|\mathbf{R}^T \mathbf{u}\|_\infty]^2, \end{aligned}$$

where (i) follows from applying (J.4), and (ii) follows from noting that by definition $\{p_c + \sum_{k=1}^{p_c} s_k(c)(1 + |\mathcal{I}_c|)\}^{1/2} \leq q_{[C]}$ and that $\mathbf{R}^T \mathbf{u}$ has more non-zero entries than $\mathbf{R}_c^T \mathbf{u}$ and contains all of its entries, i.e., $\|\mathbf{R}_c^T \mathbf{u}\|_\infty \leq \|\mathbf{R}^T \mathbf{u}\|_\infty$ for all $c \in [C]$, whence the sum is upper bounded. Thus, we arrive at (20), i.e.,

$$\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|_2 \leq 2\{\tau \mathcal{K}_{[C]}(T - p)\}^{-1/2} (Cq_{[C]})^{1/2} \|\mathbf{R}^T \mathbf{u}\|_\infty. \quad \blacksquare$$

J.4 Bounding the probability

Before proving the second part we mention a well known result for sub-Gaussian variables.

Lemma 2. *Assume that each u_i is a zero-mean sub-Gaussian variable with parameter $\sigma_i^2 < +\infty$, not necessarily independent. Let $\mathbf{u} = (u_1, \dots, u_n) \in \mathbb{R}^n$ be the vector whose i th-entry is u_i , and define $\sigma_u^2 = \max_{i \in [n]}(\sigma_i^2)$. Then, for all $\delta > 0$ we have that*

$$\Pr \left\{ \|\mathbf{u}\|_\infty > \sqrt{2}\sigma_u[\{\log(n)\}^{1/2} + \delta] \right\} \leq 2\exp(-\delta^2). \quad (\text{J.5})$$

Proof. Here, $[n] = \{1, \dots, n\}$. See Wainwright (2019, Chapter 2). \blacksquare

We are ready to prove the second part of Theorem 1, i.e., (22).

Proof. Notice that the bound given by (20) always holds once we observe a realisation of $\{\mathbf{X}_t\}$ and fix the linear model (13), i.e., it is a property of the estimator $\hat{\boldsymbol{\theta}}$ under our assumptions. Further, assume that A.3 holds. We upper bound the sub-Gaussian parameter (i.e., σ_i^2) of each entry in $\mathbf{R}^T \mathbf{u}$ as follows. Define $q = \sum_{c=1}^C \{p_c + \sum_{k=1}^{p_c} s_k(c)(1 + |\mathcal{I}_c|)\}$

$|\mathcal{I}_c|)\}$ and $\sigma_{\mathbf{u}}^2 = \max_{i \in \mathcal{K}}(\sigma_i^2)$. Notice that all j th-entries in $\mathbf{R}^T \mathbf{u}$ (i.e., $[\mathbf{R}^T \mathbf{u}]_j$ for $j \in [q]$) satisfy the following

$$\begin{aligned} [\mathbf{R}^T \mathbf{u}]_j^2 &= \langle [\mathbf{R}]_{\cdot j}, \mathbf{u} \rangle^2 \\ &\stackrel{(i)}{\leq} \|[\mathbf{R}]_{\cdot j}\|_2^2 \cdot \|\mathbf{u}\|_2^2 \\ &\stackrel{(ii)}{\leq} n^{-1} \Gamma^2 \|\mathbf{u}\|_2^2, \end{aligned}$$

where (i) follows from the Cauchy-Schwarz inequality and (ii) from assumption A.2. Thus, for all $j \in [q]$ we have that

$$\begin{aligned} E\{[\mathbf{R}^T \mathbf{u}]_j^2\} &\leq E(n^{-1} \Gamma^2 \|\mathbf{u}\|_2^2) \\ &= n^{-1} \Gamma^2 E\|\mathbf{u}\|_2^2 \\ &\leq n n^{-1} \Gamma^2 \sigma_{\mathbf{u}}^2 \\ &= \Gamma^2 \sigma_{\mathbf{u}}^2, \end{aligned} \tag{J.6}$$

above follows from applying the upper bound, taking expectations and the independence of the u_i . Thus, we see that $[\mathbf{R}^T \mathbf{u}]_j$ is sub-Gaussian with zero-mean and parameter at most $\Gamma^2 \sigma_{\mathbf{u}}^2 < +\infty$, for all $j \in [q]$.

By (J.6) we see that the variables

$$(Cq_{[C]})^{1/2} \{\tau \mathcal{K}_{[C]}(T-p)\}^{-1/2} [\mathbf{R}^T \mathbf{u}]_j,$$

are sub-Gaussian with parameter at most $\Gamma^2 \sigma_{\mathbf{u}}^2 (Cq_{[C]}) \{\tau \mathcal{K}_{[C]}(T-p)\}^{-1}$ for all $j \in [q]$.

Thus, by Lemma 2 and (J.5) we have that

$$\begin{aligned} (Cq_{[C]})^{1/2} \{\tau \mathcal{K}_{[C]}(T-p)\}^{-1/2} \|\mathbf{R}^T \mathbf{u}\|_{\infty} &\leq \\ (Cq_{[C]})^{1/2} \sigma_{\mathbf{u}} \Gamma &\left[\left\{ 2 \log \left(\sum_{c=1}^C q_c \right) \{\tau \mathcal{K}_{[C]}(T-p)\}^{-1} \right\}^{1/2} + \sqrt{2} \delta \right], \end{aligned} \tag{J.7}$$

holds with probability at least

$$1 - 2 \exp[-\delta^2 \{\tau \mathcal{K}_{[C]}(T-p)\}]. \tag{J.8}$$

Notice that the inequality between random variables given by (20), i.e.,

$$\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|_2 \leq 2 \{\tau \mathcal{K}_{[C]}(T-p)\}^{-1/2} (Cq_{[C]})^{1/2} \|\mathbf{R}^T \mathbf{u}\|_{\infty},$$

implies that

$$\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|_2 \leq 2(Cq_{[C]})^{1/2} \sigma_{\mathbf{u}} \Gamma \left[\left\{ 2 \log \left(\sum_{c=1}^C q_c \right) \{\tau \mathcal{K}_{[C]}(T-p)\}^{-1} \right\}^{1/2} + \sqrt{2} \delta \right],$$

holds with probability lower bounded by the one for (J.7). Therefore, we have that (21) holds with probability at least equal to the one in (J.8), i.e., we have shown (22). ■

K Proof of Corollary 1

We present a proof of Corollary 1. We note that the following assumption is likely a consequence of stationarity, however, we include it as an assumption for completeness and leave the proper derivation for future work.

We assume that A.1, A.2, A.3 in Assumptions 1 hold. Further, assume that as $T \rightarrow +\infty$, all three conditions hold, i.e., for all $T > 0$ there exists a $\tau > 0$ such that $\lambda_{\min}(\mathbf{R}_c^T \mathbf{R}_c) \geq \{\tau |K_c|(T - p_c)\}^{1/2}$ for all $c \in [C]$, i.e.,

$$\lim_{T \rightarrow +\infty} \lambda_{\min}(\mathbf{R}_c^T \mathbf{R}_c) \geq \lim_{T \rightarrow +\infty} \{\tau |K_c|(T - p_c)\}^{1/2} = +\infty, \quad a.s. \quad (\text{K.1})$$

for all $c \in [C]$. Further, that

$$\lim_{T \rightarrow +\infty} \max_{j \in [q]} \{||[\mathbf{R}]_{\cdot j}||_2\} \leq \lim_{T \rightarrow +\infty} n^{-1/2} \Gamma = 0,$$

where $n := \min_{c \in [C]} \{|K_c|(T - p_c)\}$, i.e., the number of rows in $\mathbf{R}$.

K.1 Proof of consistency

We proceed to prove consistency, i.e., (23).

Proof. By (K.1), our assumptions and the above means that

$$\lim_{T \rightarrow \infty} 2\sigma_u^2 \Gamma^2 \{Cq_{[C]} \log(Cq_{[C]})\} \{\tau \mathcal{K}_{[C]}(T - p)\}^{-1} = 0. \quad (\text{K.2})$$

By (22), for all $T > 0$ and $\delta > 0$, we have that

$$\begin{aligned} 0 \leq \mathbf{Pr} \left(||\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0||_2 > 2(Cq_{[C]})^{1/2} \sigma_u \Gamma \left[\left\{ 2 \log \left(\sum_{c=1}^C q_c \right) \{\tau \mathcal{K}_{[C]}(T - p)\}^{-1} \right\}^{1/2} + \delta \right] \right) \\ \leq 2 \exp \left[-\frac{\delta^2}{2} \{\tau \mathcal{K}_{[C]}(T - p)\} \right], \quad (\text{K.3}) \end{aligned}$$

hence, by applying (K.1), (K.2) and the squeeze theorem to (K.3) we arrive at (23), i.e., as $T \rightarrow \infty$ we have that $||\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0||_2 \xrightarrow{\mathbf{Pr}} 0$. ■

K.2 Asymptotic normality

We introduce a general notation for random variables and M-estimators below. This is required for presenting the general results for asymptotic normality in K.2.2. These are adapted into our GNAR problem in K.3 for proving asymptotic normality (Corollary 1). Readers familiar with M-estimators and empirical processes may go to K.3. Throughout this subsection we use the following definitions.

K.2.1 Preliminary definitions

Let $\mathbf{x} \in \mathcal{X}$ be a random variable with distribution function $F_{\mathbf{x}}$, and $g_{\boldsymbol{\theta}}$ be a loss function indexed by $\boldsymbol{\theta} \in \boldsymbol{\Theta} \subseteq \mathbb{R}^q$. Assume that we observe n realisations of $\mathbf{x}$, which are independent and identically distributed. Define the following.

- $F_{\mathbf{x}}g_{\boldsymbol{\theta}} := E_{\mathbf{x}}\{g_{\boldsymbol{\theta}}(\mathbf{x})\}$, i.e., the expected value of the loss function,
- $\hat{F}_{\mathbf{x}}g_{\boldsymbol{\theta}} := n^{-1} \sum_{i=1}^n g_{\boldsymbol{\theta}}(\mathbf{x}_i)$, i.e., the empirical mean (average) of the loss function for n observations,
- $\dot{g}_{\boldsymbol{\theta}} := \dot{g}_{\boldsymbol{\theta}}(\mathbf{x}) = \nabla_{\boldsymbol{\theta}}g_{\boldsymbol{\theta}}(\mathbf{x})$, i.e., the gradient with respect to $\boldsymbol{\theta}$ of the loss function with $\mathbf{x}$ fixed.

K.2.2 Conditions for asymptotic normality

Our proof of asymptotic normality relies on arguments from empirical process theory and M-estimators. The conditions below are taken from Chapter 10 in van de Geer (2020). The proof of asymptotic normality for M-estimators relies on asymptotic continuity, consistency and exploits Vapnik-Chervonenkis theory. See Chapters 7 to 10 in van de Geer (2020).

1. There exists an $\epsilon > 0$ such that for all $\boldsymbol{\theta}$ that satisfy $\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|_2 < \epsilon$, the function $g_{\boldsymbol{\theta}}$ is differentiable with gradient $\dot{g}_{\boldsymbol{\theta}}$ for all $x \in \mathcal{X}$, i.e., all partial derivatives exist in an ℓ_2 -norm neighbourhood of $\boldsymbol{\theta}_0$. See Condition (a) in Chapter 10.2 of van de Geer (2020).
2. As $\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}_0$, we have that

$$F_{\mathbf{x}}(\dot{g}_{\boldsymbol{\theta}} - \dot{g}_{\boldsymbol{\theta}_0}) = \mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\theta}_0) + o(\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|),$$

where $\mathbf{H} \in \mathbb{R}^{q \times q}$ is a symmetric positive definite matrix, i.e., the second order approximation is well defined as $\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}_0$. See Condition (b) in Chapter 10.2 of van de Geer (2020).

3. The ℓ_2 -norm of gradient differences is continuous, i.e.,

$$\lim_{\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}_0} \|\dot{g}_{\boldsymbol{\theta}} - \dot{g}_{\boldsymbol{\theta}_0}\|_2 = 0.$$

See Condition (c.2) in Chapter 10.2 of van de Geer (2020).

4. There exists an $\epsilon > 0$ such that the function class $\mathfrak{G}_{\epsilon} := \{\dot{g}_{\boldsymbol{\theta}} : \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|_{\infty} < \epsilon\}$ is asymptotically continuous at $\dot{g}_{\boldsymbol{\theta}_0}$, i.e., as $n \rightarrow +\infty$, if the random sequence $\{\dot{g}_{\boldsymbol{\theta}_n}\} \subset \mathfrak{G}_{\epsilon}$ satisfies $F_{\mathbf{x}}(\|\dot{g}_{\boldsymbol{\theta}_n} - \dot{g}_{\boldsymbol{\theta}_0}\|_2^2) \xrightarrow{\text{Pr}} 0$, then

$$|z_n(\boldsymbol{\theta}_n) - z_n(\boldsymbol{\theta}_0)| \xrightarrow{\text{Pr}} 0,$$

where $z_n(\boldsymbol{\theta}) := \sqrt{n}(F_{\mathbf{x}} - \hat{F}_{\mathbf{x}})\dot{g}_{\boldsymbol{\theta}}$ is the empirical process indexed by the class $\mathfrak{G}_{\epsilon}$ for the distribution $F_{\mathbf{x}}$. That is, the function class $\mathfrak{G}_{\epsilon}$ is such that approximating the expected value of the idealised (theoretical) loss by the empirical one in a neighbourhood of $\boldsymbol{\theta}_0$ is valid. See Condition (c.1) in Chapter 10.2 of van de Geer (2020).

K.2.3 Preliminary results for showing asymptotic normality

Theorem 5. Assume that $\mathfrak{G}_\theta := \{g_\theta : \theta \in \Theta\}$ is a Vapnik-Chervonenkis graph class with envelope function $G_\theta := \sup_{g \in \mathfrak{G}_\theta} \{|g_\theta(\mathbf{x})|\}$, $\mathbf{x} \in \mathcal{X}$, such that $G_\theta \in L_2(F_{\mathbf{x}})$, i.e., $F_{\mathbf{x}} G_\theta^2 < +\infty$,

and that all (possibly random) sequences $\{g_{\theta_n}\} \subset \mathfrak{G}_\theta$ satisfy $\|g_{\theta_n} - g_{\theta_0}\|_2 \xrightarrow{\text{Pr}} 0$ as $\theta_n \xrightarrow{\text{Pr}} \theta_0$, i.e., the function class is mean square continuous with respect to θ_0 . Then,

$$|z_n(\theta_n) - z_n(\theta_0)| \xrightarrow{\text{Pr}} 0,$$

i.e., the class $\mathfrak{G}_\theta$ is asymptotically continuous at g_{θ_0} .

Proof. See Chapter 9 in van de Geer (2020), and Chapter 6 in van de Geer (2020) and Chapter 4 in Wainwright (2019) for the definition and properties of a Vapnik-Chervonenkis graph class and Vapnik-Chervonenkis dimension. ■

Theorem 6. Assume that $\hat{\theta}$ is a consistent M-estimator of $\theta_0 \in \Theta \subseteq \mathbb{R}^q$, where θ_0 is an interior point. Further, assume that conditions 1, 2, 3 and 4 in Conditions K.2.2 hold. Then, we have that

$$\sqrt{n}(\hat{\theta} - \theta_0) = -\hat{F}(\mathbf{H}^{-1}\dot{g}_{\theta_0}) + o_{\text{Pr}}(1),$$

i.e., $\hat{\theta}$ is asymptotically linear with influence function $-\mathbf{H}^{-1}\dot{g}_{\theta_0}$. Thus, we have that

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N_q(0, \Sigma_{\hat{\theta}}),$$

where $\Sigma_{\hat{\theta}} = \mathbf{H}^{-1}\{F(\dot{g}_{\theta_0}\dot{g}_{\theta_0}^T)\}\mathbf{H}^{-1}$. That is, $\hat{\theta}$ is asymptotically normal with covariance given by the local behaviour of the loss function around θ_0 .

Proof. See Theorem 5.21 in van der Vaart (2008, Chapter 5) and van de Geer (2020, Chapter 10). ■

K.3 Proof of asymptotic normality

Before proving asymptotic normality we present a summary of the proof.

1. Show that the problem is equivalent to analysing $n := T - p \in \mathbb{Z}^*$ i.i.d. observations of a d -dimensional linear regression problem with $q < nd$ (usually $q \ll nd$) unknown parameters;
2. Show that the q leading singular values of the expected value of the design matrix are positive and finite, i.e., the *idealised* (expected) linear regression problem has a unique minimum and well defined Hessian;
3. Show that $\hat{\theta}$ is an M-estimator, and that θ_0 minimises the idealised loss and is an interior point;
4. Apply Theorem 1 for showing that $\hat{\theta}$ is consistent, i.e., applying (23);
5. Show that conditions 1, 2, 3 and 4 in Conditions K.2.2 hold;
6. Apply Lemma 6 for finishing the proof.

K.3.1 Problem setting

Assume Assumptions 1, (K.1) and (K.2) hold and that we observe a realisation of length T of a stationary GNAR(p) (maximum lag is p) process $\mathbf{X}_t \in \mathbb{R}^d$ with q unknown parameters. Let $t_i := p + i$ for $i \in [n]$, where $n = T - p$. Also, $\mathbf{y}_{t_i} := \mathbf{X}_{p+i}$ for all $t = p + 1, \dots, T$. Then, we can write (8) as

$$\mathbf{y}_{t_i} = \mathbf{Z}_{t_i} \boldsymbol{\theta}_0 + \mathbf{u}_{t_i}, \quad (\text{K.4})$$

where $\mathbf{Z}_{t_i} \in \mathbb{R}^{d \times q}$ is the design matrix with p previous lags and the corresponding within and between community components, i.e.,

$$\begin{aligned} \mathbf{R}_{k,t,c} &:= \left[\mathbf{X}_{t-k}^c | \mathbf{Z}_{t-k}^{1,c} | \dots | \mathbf{Z}_{t-k}^{s_k(c),c} | \mathbf{Z}_{t-k}^{1,c;\tilde{c}} | \dots | \mathbf{Z}_{t-k}^{s_k(c),c;\tilde{c}} \right], \\ \mathbf{R}_{k,t} &:= [\mathbf{R}_{k,t,c} | \dots | \mathbf{R}_{k,t,C}], \\ \mathbf{Z}_{t_i} &:= [\mathbf{R}_{1,t} | \dots | \mathbf{R}_{p,t}], \end{aligned} \quad (\text{K.5})$$

where each predictor column for $\tilde{c} \in \mathcal{I}_c$ is concatenated in ascending order with respect to $\tilde{c}$, i.e., if $\tilde{c}_2 > \tilde{c}_1$, then the column for $\tilde{c}_1$ precedes the one for $\tilde{c}_2$, $\mathbf{u}_{t_i}$ is a sub-Gaussian multivariate white noise process with a diagonal covariance, and $\boldsymbol{\theta}_0 \in \boldsymbol{\Theta} \subset \mathbb{R}^q$ is the “true” vector of parameters.

Let $F_{\mathbf{y}_t, \mathbf{Z}_t}$ denote the joint distribution function, $F_{\mathbf{y}_t | \mathbf{Z}_t}$ denote the conditional distribution function, and $F_{\mathbf{Z}_t}$, $F_{\mathbf{u}}$ denote the marginal distribution functions. Then, for any function $g_{\boldsymbol{\theta}}$ we have that

$$\begin{aligned} F_{\mathbf{y}_t, \mathbf{Z}_t}(g_{\boldsymbol{\theta}}) &= F_{\mathbf{Z}_t}\{F_{\mathbf{y}_t | \mathbf{Z}_t}(g_{\boldsymbol{\theta}})\}, \\ F_{\mathbf{u}, \mathbf{Z}_t}(g_{\boldsymbol{\theta}}) &= F_{\mathbf{Z}_t}\{F_{\mathbf{u}}(g_{\boldsymbol{\theta}})\}, \end{aligned}$$

i.e., we can compute the relevant risks by iterated expectations.

Notice that by (K.4) we can factorise the joint density of the data by exploiting that $\mathbf{u}_{t_i}$ are *i.i.d.* and independent of $\mathbf{Z}_{t_i}$ for all $i \in [n]$. Hence, we have that

$$f(\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \boldsymbol{\theta}_0) = f(\mathbf{u}_{t_i}),$$

thus, by exploiting the Markov property above for t_i , using stationarity of $\mathbf{Z}_{t_i}$ for each t_i , and letting $(\mathbf{y}, \mathbf{Z}) \in \mathbb{R}^{nd} \times \mathbb{R}^{nd \times q}$ denote the sample, we have that

$$f(\mathbf{y}, \mathbf{Z}) = f(\mathbf{Z}_t) \prod_{i=1}^n f_{\mathbf{u}}(\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \boldsymbol{\theta}_0),$$

thus, we see that there are n *i.i.d.* observations of the linear regression problem (K.4), albeit we still have to account for the first p time-steps, i.e., $f(\mathbf{Z}_{t_1})$. We circumvent this by performing iterated expectations with respect to the n linear regression problems. We finish the problem setting with Proposition 2.

Proposition 2. *Assume that each stationary process $[\mathbf{Z}_t]_{ij} \neq 0$ in $\mathbf{Z}_t \in \mathbb{R}^{d \times q}$, where $q_c \leq d_c$ for all $c \in [C]$, i.e., lagged entries of the same process for each community, has a spectral density $f_k(\omega)$ such that there exist two constants $0 < m \leq M < +\infty$, which satisfy*

$$m \leq f_{k_i k_j}(\omega) \leq M,$$

for all $\omega \in [-\pi, \pi]$. Then, we have that

$$0 < 2\pi md \leq \lambda_{\min} [E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}] \leq \lambda_{\max} [E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}] \leq 2d\pi M < +\infty, \quad (\text{K.6})$$

i.e., all eigenvalues of $E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}$ are positive and finite. Further, the matrix $E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}$ is symmetric positive definite.

Proof. Notice that all entries in $(\mathbf{Z}_t^T \mathbf{Z}_t)$ correspond to the sum of products of either auto-covariances between the same entries in $\mathbf{Z}_t$ or covariances between different entries in $\mathbf{Z}_t$ at lags $h = 0, \dots, p-1$, i.e, for $(i, j) \in [q] \times [q]$ we have that

$$[(\mathbf{Z}_t^T \mathbf{Z}_t)]_{ij} = \sum_{k=1}^d [\mathbf{Z}_t]_{ki} [\mathbf{Z}_t]_{kj},$$

where the columns i and j will vary between covariances and autocovariances at different lags for autoregressive, between and within community columns in $\mathbf{Z}_t$ for the k th stationary process with a maximum lag equal to $p-1$. Define $\mathbf{\Gamma}_k(p, [C]) \in \mathbb{R}^{q \times q}$ as the matrix with entries

$$[\mathbf{\Gamma}_k(p, [C])]_{ij} := E\{[\mathbf{Z}_t]_{ki} [\mathbf{Z}_t]_{kj}\},$$

Thus, we have that

$$E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\} = \sum_{k=1}^d \mathbf{\Gamma}_k(p, [C]). \quad (\text{K.7})$$

Notice that by assumption we have that the spectral density for each k th process and the cross-spectral density between two correlated processes for all $k \in \mathcal{K}$ and all $\omega \in [-\pi, \pi]$ satisfy

$$0 < m \leq f_{k_i}(\omega) \leq M < +\infty, \quad (\text{K.8})$$

$$|f_{k_i, k_j}(\omega)| \leq \{f_{k_i}(\omega) f_{k_j}(\omega)\}^{1/2}, \quad (\text{K.9})$$

where k_i denotes the i th process linked to node k (e.g., number of within community r -stage neighbourhood regressions). Further, if the cross spectral density is non-zero, then

$$0 < m \leq \text{Re}\{f_{k_i, k_j}(\omega)\}, \quad (\text{K.10})$$

$$0 < m \leq \text{Im}\{f_{k_i, k_j}(\omega)\}. \quad (\text{K.11})$$

Let $\gamma_{k,ij}(h) = [\mathbf{\Gamma}_k(p, [C])]_{ij}$. Then, it is the (auto)covariance of the i th and j th columns for the k th row in $\mathbf{Z}_t$ at lag $h = 0, \dots, p-1$, i.e,

$$\gamma_{k,ij}(h) = \begin{cases} E(X_{k,t+h} X_{k,t}), & \text{if } i, j \text{ are s.t. } k \text{ is autoregressive,} \\ E(X_{k,t+h} Z_{k,t}^{r,c}), & \text{if } i, j \text{ are s.t. } k \text{ is within,} \\ E(X_{k,t+h} Z_{k,t}^{r,c;\tilde{c}}), & \text{if } i, j \text{ are s.t. } k \text{ is interaction,} \\ E(Z_{k,t+h}^{r,c} Z_{k,t}^{r,c}), & \text{if } i, j \text{ are s.t. } k \text{ is cross-within,} \\ E(Z_{k,t+h}^{r,c;\tilde{c}} Z_{k,t}^{r,c;\tilde{c}}), & \text{if } i, j \text{ are s.t. } k \text{ is cross-interaction,} \\ E(Z_{k,t+h}^{r,c} Z_{k,t}^{r,c;\tilde{c}}), & \text{if } i, j \text{ are s.t. } k \text{ is within-interaction.} \end{cases} \quad (\text{K.12})$$

Since all k_i -th processes are stationary, by the spectral representation of the autocovariance function; see Brockwell & Davis (2006), we have that

$$\gamma_{k,ij}(h) = \int_{-\pi}^{\pi} f_{k_i,k_j}(\omega) \exp(\imath\pi h\omega) d\omega, \quad (\text{K.13})$$

where $\imath$ is the imaginary unit (i.e., $\imath^2 = -1$), and $f_{k_i,k_i}(\omega) = f_{k_i}(\omega)$ is the (cross)spectral density of the k_i -th process.

Let $\mathbf{a} \in \mathbb{R}^q \setminus \{\mathbf{0}\}$ be an eigenvector of $\mathbf{\Gamma}_k(p, [C])$ with eigenvalue $\lambda_k \neq 0$. Then, we have that

$$\begin{aligned} \lambda_k \|\mathbf{a}\|_2^2 &= \langle \mathbf{a}, \mathbf{\Gamma}_k(p, [C]) \mathbf{a} \rangle \\ &= \sum_{i=1}^q \sum_{j=1}^q a_i a_j \gamma_{k,ij}(h) \end{aligned}$$

- By applying (K.13) we have that

$$\begin{aligned} \lambda_k \|\mathbf{a}\|_2^2 &= \sum_{i=1}^q \sum_{j=1}^q \int_{-\pi}^{\pi} f_{k_i,k_j}(\omega) \exp(\imath\pi h\omega) d\omega \\ &\leq \int_{-\pi}^{\pi} \sum_{i=1}^q \sum_{j=1}^q a_i a_j |f_{k_i,k_j}(\omega)| \exp(\imath\pi h\omega) d\omega \end{aligned}$$

- By applying (K.9) we have that

$$\lambda_k \|\mathbf{a}\|_2^2 \leq \int_{-\pi}^{\pi} \sum_{i=1}^q \sum_{j=1}^q a_i a_j \{f_{k_i}(\omega) f_{k_j}(\omega)\}^{1/2} \exp(\imath\pi h\omega) d\omega$$

- By applying (K.8) we have that

$$\lambda_k \|\mathbf{a}\|_2^2 \leq \int_{-\pi}^{\pi} \sum_{i=1}^q \sum_{j=1}^q a_i a_j M \exp(\imath\pi h\omega) d\omega$$

- By noting that $\int_{-\pi}^{\pi} \exp(\imath\pi h\omega) d\omega = \mathbb{I}(h=0)$ we have that

$$\begin{aligned} \lambda_k \|\mathbf{a}\|_2^2 &= M \sum_{i=1}^q \sum_{j=1}^q \int_{-\pi}^{\pi} a_i a_j \mathbb{I}(i=j) d\omega \\ &= 2\pi M \|\mathbf{a}\|_2^2 < +\infty, \end{aligned}$$

where $\mathbb{I}(i=j)$ indicates one of the cases in (K.12) with $h=0$. Similarly, we have that

$$\begin{aligned} \lambda_k \|\mathbf{a}\|_2^2 &= \langle \mathbf{a}, \mathbf{\Gamma}_k(p, [C]) \mathbf{a} \rangle \\ &= \sum_{i=1}^q \sum_{j=1}^q a_i a_j \gamma_{k,ij}(h) \end{aligned}$$

- By applying (K.13) we have that

$$\begin{aligned}\lambda_k ||\mathbf{a}||_2^2 &= \sum_{i=1}^q \sum_{j=1}^q \int_{-\pi}^{\pi} a_i a_j \exp(i\pi h\omega) f_{k_i, k_j}(\omega) d\omega \\ &= \sum_{i=1}^q \sum_{j=1}^q \int_{-\pi}^{\pi} a_i a_j \exp(i\pi h\omega) [\operatorname{Re}\{f_{k_i, k_j}(\omega)\} + i\operatorname{Im}\{f_{k_i, k_j}(\omega)\}] d\omega\end{aligned}$$

- By noting that the left hand side is a real number, we see that

$$i \sum_{i=1}^q \sum_{j=1}^q \int_{-\pi}^{\pi} a_i a_j \exp(i\pi h\omega) \operatorname{Im}\{f_{k_i, k_j}(\omega)\} d\omega = 0,$$

and by applying (K.10) we have that

$$\begin{aligned}\lambda_k ||\mathbf{a}||_2^2 &= \sum_{i=1}^q \sum_{j=1}^q \int_{-\pi}^{\pi} a_i a_j \exp(i\pi h\omega) \operatorname{Re}\{f_{k_i, k_j}(\omega)\} d\omega \\ &\geq m \int_{-\pi}^{\pi} \sum_{i=1}^q \sum_{j=1}^q a_i a_j \exp(i\pi h\omega) d\omega \\ &= m \sum_{i=1}^q \sum_{j=1}^q \int_{-\pi}^{\pi} a_i a_j \mathbb{I}(i=j) d\omega \\ &= 2\pi m ||\mathbf{a}||_2^2 > 0.\end{aligned}$$

Thus, we have shown that any eigenvalue $\lambda_k \neq 0$ of $\mathbf{\Gamma}_k(p, [C])$ satisfies

$$0 < 2\pi m ||\mathbf{a}||_2^2 \leq \lambda_k ||\mathbf{a}||_2^2 = \langle \mathbf{a}, \mathbf{\Gamma}_k(p, [C]) \mathbf{a} \rangle \leq 2\pi M ||\mathbf{a}||_2^2 < +\infty. \quad (\text{K.14})$$

Let $\mathbf{a} \in \mathbb{R}^q \setminus \{\mathbf{0}\}$ be an eigenvector of $\mathbf{\Gamma}(p, [C]) := E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}$ with eigenvalue $\lambda \neq 0$. Then, we have that

$$\begin{aligned}\lambda ||\mathbf{a}||_2^2 &= \langle \mathbf{a}, \mathbf{\Gamma}(p, [C]) \mathbf{a} \rangle \\ &= \langle \mathbf{a}, \sum_{k=1}^d \mathbf{\Gamma}_k(p, [C]) \mathbf{a} \rangle \\ &= \sum_{k=1}^d \langle \mathbf{a}, \mathbf{\Gamma}_k(p, [C]) \mathbf{a} \rangle,\end{aligned}$$

and notice that by (K.14) we have that

$$0 < \sum_{k=1}^d 2\pi m ||\mathbf{a}||_2^2 \leq \lambda ||\mathbf{a}||_2^2 \leq \sum_{k=1}^d 2\pi M ||\mathbf{a}||_2^2 < +\infty,$$

whence we arrive at (K.6), i.e.,

$$0 < 2\pi m d \leq \lambda_{\min} [E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}] \leq \lambda_{\max} [E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}] \leq 2d\pi M < +\infty,$$

which implies that $E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}$ is positive definite, and by symmetry of the autocovariance function of stationary processes and (K.12) it is also symmetric. $\blacksquare$

Assuming that $q_c \leq d_c$ can be relaxed by noting that by stacking m time-steps coming from (K.4), such that $q_c \leq md_c$ for all $c \in [C]$. Since the m white noise vectors $\mathbf{u}_t$ are independent, augmenting the design in this way results in a setting where (K.4) is not underdetermined (i.e., $q \leq md$).

Proposition 3. *Let $\hat{\boldsymbol{\theta}}$ be given by (14) and $\boldsymbol{\theta}_0$ be the “true” vector of parameters, both in $\boldsymbol{\Theta} \subset \mathbb{R}^q$, where $\boldsymbol{\Theta}$ is the open set which satisfies the stationarity constraints given by Corollary 2. Define the squared error loss function $g_{\boldsymbol{\theta}}$ for (K.4) and all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ as*

$$g_{\boldsymbol{\theta}} : \boldsymbol{\Theta} \longrightarrow \mathbb{R}_0^+ \\ \boldsymbol{\theta} \mapsto (2d)^{-1} \|\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \boldsymbol{\theta}\|_2^2. \quad (\text{K.15})$$

Then, we have that

$$\boldsymbol{\theta}_0 = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \{F_{\mathbf{y}_t, \mathbf{Z}_t}(g_{\boldsymbol{\theta}})\}, \quad (\text{K.16})$$

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \{\hat{F}_{\mathbf{y}_t | \mathbf{Z}_t}(g_{\boldsymbol{\theta}})\}, \quad (\text{K.17})$$

i.e., $\hat{\boldsymbol{\theta}}$ is the M-estimator under $g_{\boldsymbol{\theta}}$ with idealised value $\boldsymbol{\theta}_0$ in our problem setting, where $\boldsymbol{\theta}_0$ is an interior point of $\boldsymbol{\Theta}$ that minimises the idealised loss. Further, we have that the M-estimator is consistent, i.e., as $T \rightarrow \infty$ we have that $\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|_2 \xrightarrow{\text{Pr}} 0$.

Proof. We start by analysing the idealised loss. Notice that

$$\begin{aligned} F_{\mathbf{y}_t | \mathbf{Z}_t}(g_{\boldsymbol{\theta}}) &= F_{\mathbf{y}_t | \mathbf{Z}_t} (2d)^{-1} \|\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \boldsymbol{\theta}\|_2^2 \\ &= F_{\mathbf{y}_t | \mathbf{Z}_t} (2d)^{-1} \|\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \boldsymbol{\theta} + \mathbf{Z}_{t_i} \boldsymbol{\theta}_0 - \mathbf{Z}_{t_i} \boldsymbol{\theta}_0\|_2^2 \\ &= (2d)^{-1} F_{\mathbf{u}_t} \|\mathbf{u}_{t_i} - \mathbf{Z}_{t_i}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)\|_2^2 \\ &= (2d)^{-1} \{F_{\mathbf{u}_t} \|\mathbf{u}_{t_i}\|_2^2 + F_{\mathbf{u}_t} \|\mathbf{Z}_{t_i}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)\|_2^2\} \\ &= (2d)^{-1} \{\operatorname{tr}(\boldsymbol{\Sigma}_{\mathbf{u}}) + \|\mathbf{Z}_{t_i}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)\|_2^2\}. \end{aligned}$$

Notice that the above is the sum of two non-negative numbers, thus, it is minimised if $\boldsymbol{\theta} = \boldsymbol{\theta}_0$, i.e., $\boldsymbol{\theta}_0$ is an idealised minimiser of $g_{\boldsymbol{\theta}}$.

By Proposition 2 the risk is finite, and notice that

$$0 \leq E \|\mathbf{Z}_{t_i}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)\|_2^2 \leq \tau_{(q)} \|(\boldsymbol{\theta} - \boldsymbol{\theta}_0)\|_2^2,$$

thus, by iterated expectation we have that

$$F_{\mathbf{y}_t, \mathbf{Z}_t}(g_{\boldsymbol{\theta}_0}) \leq F_{\mathbf{y}_t, \mathbf{Z}_t}(g_{\boldsymbol{\theta}}) \leq \sigma_{\mathbf{u}}^2 + \tau_{(q)} \|(\boldsymbol{\theta} - \boldsymbol{\theta}_0)\|_2^2 < +\infty, \quad (\text{K.18})$$

where $\tau_{(q)} := \lambda_{\max} [E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}]$.

Notice that squared error loss is differentiable for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ and all $(\mathbf{y}_t, \mathbf{Z}_t) \in \mathbb{R}^d \times \mathbb{R}^{d \times q}$ because it is the composition of differentiable maps. Thus, we have that

$$\dot{g}_{\boldsymbol{\theta}} = -d^{-1} \{\mathbf{Z}_{t_i}^T (\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \boldsymbol{\theta})\}, \quad (\text{K.19})$$

for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ and all $(\mathbf{y}_t, \mathbf{Z}_t) \in \mathbb{R}^d \times \mathbb{R}^{d \times q}$.

By Proposition 2 we have that the matrix $E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}$ is non-singular. Thus, the risk is

$$\begin{aligned} F_{\mathbf{y}_t, \mathbf{Z}_t}(\dot{g}_\theta) &= -d^{-1} F_{\mathbf{y}_t, \mathbf{Z}_t} \{ \mathbf{Z}_{t_i}^T (\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \theta) \} \\ &= -d^{-1} [F_{\mathbf{y}_t, \mathbf{Z}_t} \mathbf{Z}_{t_i}^T \{ \mathbf{u}_{t_i} - \mathbf{Z}_{t_i} (\theta - \theta_0) \}] , \end{aligned}$$

and by iterated expectation we have that

$$F_{\mathbf{y}_t, \mathbf{Z}_t}(\dot{g}_\theta) = d^{-1} [-F_{\mathbf{Z}_t} \mathbf{Z}_{t_i}^T F_{\mathbf{u}_t}(\mathbf{u}_{t_i}) + F_{\mathbf{Z}_t}(\mathbf{Z}_t^T \mathbf{Z}_t)(\theta - \theta_0)] , \quad (\text{K.20})$$

since $E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}$ is symmetric positive definite and non-singular, and $F_{\mathbf{u}_t}(\mathbf{u}_{t_i}) = \mathbf{0}$, above equals zero if and only if $\theta = \theta_0$. Moreover, the Hessian of g_θ is

$$\nabla_{\theta}^2 g_\theta = d^{-1} (\mathbf{Z}_t^T \mathbf{Z}_t) , \quad (\text{K.21})$$

for all $\theta \in \Theta$ and all $(\mathbf{y}_t, \mathbf{Z}_t) \in \mathbb{R}^d \times \mathbb{R}^{d \times q}$. Hence, by noting that the expectation of $\nabla_{\theta}^2 g_\theta$ given by (K.21) is a positive definite Hessian matrix, we have that θ_0 is the idealised global minima of the convex loss function (K.15), i.e., we have shown (K.16). Thus, we can solve for θ_0 by integrating (K.19) the result to zero. Thus, we have the well known result from linear regression and least-squares for the conditional problem

$$\theta_0 = E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}^{-1} E(\mathbf{Z}_t^T \mathbf{y}_t) ,$$

which, by noticing that the expectations are autocovariances, can be written in a Yule-Walker like way

$$E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\} \theta_0 = E(\mathbf{Z}_t^T \mathbf{y}_t) .$$

We proceed with the empirical risk minimiser. Note that when we concatenate all $\mathbf{Z}_{t_i}$ for all time steps t_i by rows, i.e., each matrix is placed below the preceding one according to the t_i . Then, we have that the resulting matrix is equal to $\mathbf{R}$ in Section 3, hence, we can write

$$\begin{aligned} \hat{\theta} &= \underset{\theta}{\operatorname{argmin}} \left\{ \hat{F}_{\mathbf{y}_t | \mathbf{Z}_t}(g_\theta) \right\} , \\ &= \underset{\theta}{\operatorname{argmin}} (2d)^{-1} \sum_{i=1}^n \|\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \theta\|_2^2 \\ &= \underset{\theta}{\operatorname{argmin}} (2d)^{-1} \|\mathbf{y} - \mathbf{R} \theta\|_2^2 , \\ &\stackrel{(i)}{=} \underset{\theta}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{R} \theta\|_2^2 , \end{aligned}$$

where (i) follows by noting that $(2d)^{-1}$ is a positive constant that does not affect the optimal value, and $(\mathbf{y}, \mathbf{R})$ are the same ones as in (13) in Section 3, thus, we have that

$$\hat{\theta} = (\mathbf{R}^T \mathbf{R})^{-1} \mathbf{R}^T \mathbf{y} ,$$

i.e., the M-estimator is the (conditional) least-squares estimator defined in Section 3, i.e., we have shown (K.17).

Finally, by Theorem 1 and the first part of Corollary 1 we have that the M-estimator is consistent, i.e., as $T \rightarrow \infty$ we have that $\|\hat{\theta} - \theta_0\|_2 \xrightarrow{\text{Pr}} 0$. ■

Proposition 4. Let g_θ be given by (K.15) in Proposition 3. Then, there exists an $\epsilon > 0$ such that

$$F_{\mathbf{y}_t, \mathbf{Z}_t}(\dot{g}_\theta - \dot{g}_{\theta_0}) = d^{-1} E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}(\theta - \theta_0) + o(\|\theta - \theta_0\|), \quad (\text{K.22})$$

for all θ such that $\|\theta - \theta_0\| < \epsilon$, i.e., the loss function and problem setting satisfy Condition 2 in Conditions K.2.2.

Proof. Notice that by (K.21) and (K.19) for an $\epsilon > 0$ by Taylor's Theorem for multivariate functions for all $\theta \in \Theta$ such that $\|\theta - \theta_0\|_2 < \epsilon$, we have that

$$g_\theta - g_{\theta_0} = \langle \dot{g}_{\theta_0}, (\theta - \theta_0) \rangle + 2^{-1}(\theta - \theta_0)^T (\mathbf{Z}_t^T \mathbf{Z}_t)(\theta - \theta_0) + o(1) \|\theta - \theta_0\|_2^2, \quad (\text{K.23})$$

where $o(\|\theta - \theta_0\|_2^2) \rightarrow 0$ as $\epsilon \rightarrow 0$.

Hence, by taking iterated expectations in (K.23) and using (K.20), i.e., $F_{\mathbf{y}_t, \mathbf{Z}_t}(\dot{g}_{\theta_0}) = 0$, we have that

$$F_{\mathbf{y}_t, \mathbf{Z}_t}(g_\theta - g_{\theta_0}) = 2^{-1}(\theta - \theta_0)^T E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}(\theta - \theta_0) + o(1) \|\theta - \theta_0\|_2^2. \quad (\text{K.24})$$

Notice that θ_0 is an interior point and that by (K.18) the integral is bounded for all θ , thus, we can differentiate both sides of (K.24) to get

$$F_{\mathbf{y}_t, \mathbf{Z}_t}(\dot{g}_\theta - \dot{g}_{\theta_0}) = E\{(\mathbf{Z}_t^T \mathbf{Z}_t)\}(\theta - \theta_0) + o(1) \|\theta - \theta_0\|_2,$$

i.e., we have shown (K.22). Alternatively, we can look at (K.20) and notice that the error is accounted for by the smoothness and nice properties of squared error loss. $\blacksquare$

Proposition 5. Let g_θ be given by (K.15) in Proposition 3 in the same problem setting. Then, we have that

$$\lim_{\theta \rightarrow \theta_0} \|\dot{g}_\theta - \dot{g}_{\theta_0}\|_2 = 0, \quad (\text{K.25})$$

i.e., the loss function g_θ is mean squared continuous.

Proof. Notice that

$$\begin{aligned} \|\dot{g}_\theta - \dot{g}_{\theta_0}\|_2^2 &= \langle \dot{g}_\theta - \dot{g}_{\theta_0}, \dot{g}_\theta - \dot{g}_{\theta_0} \rangle \\ &= d^{-2} \left\langle \mathbf{Z}_{t_i}^T (\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \theta) - \mathbf{Z}_{t_i}^T (\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \theta_0), \right. \\ &\quad \left. \mathbf{Z}_{t_i}^T (\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \theta) - \mathbf{Z}_{t_i}^T (\mathbf{y}_{t_i} - \mathbf{Z}_{t_i} \theta_0) \right\rangle \\ &= d^{-2} \langle (\mathbf{Z}_t^T \mathbf{Z}_t)(\theta - \theta_0), (\mathbf{Z}_t^T \mathbf{Z}_t)(\theta - \theta_0) \rangle \\ &= d^{-2} \|(\mathbf{Z}_t^T \mathbf{Z}_t)(\theta - \theta_0)\|_2^2. \end{aligned}$$

- Thus, as $\theta - \theta_0 \rightarrow 0$, the above goes to zero, i.e.,

$$\lim_{\theta \rightarrow \theta_0} \|\dot{g}_\theta - \dot{g}_{\theta_0}\|_2^2 = 0$$

- By continuity of the square root function we arrive at (K.25), i.e.,

$$\lim_{\theta \rightarrow \theta_0} \|\dot{g}_\theta - \dot{g}_{\theta_0}\|_2 = 0.$$

- Further, by linearity of expectation and Proposition 2 we also have that

$$F_{\mathbf{y}_t, \mathbf{Z}_t} \|\dot{g}_\theta - \dot{g}_{\theta_0}\|_2 \leq d^{-1} \tau_{(q)} \|\theta - \theta_0\|_2.$$

$\blacksquare$

Proposition 6. Let $\mathfrak{G}_\epsilon := \{\dot{g}_\theta : \|\theta - \theta_0\|_2 < \epsilon\}$ for some $\epsilon > 0$, and define the envelope function $G_\epsilon := \sup_{\dot{g}_\theta \in \mathfrak{G}_\epsilon} \{|\dot{g}_\theta(\mathbf{y}_{t_i}, \mathbf{Z}_{t_i})|_j\}$ for $j \in [q]$, i.e., the absolute value supremum among all entries in $\dot{g}_\theta$. Then, we have that

1. The class $\mathfrak{G}_\epsilon$ is a finite dimension Vapnik-Chervonenkis graph class.
2. $G_\epsilon \in L_2(F_{\mathbf{y}_t, \mathbf{Z}_t})$, i.e., $F_{\mathbf{y}_t, \mathbf{Z}_t}[G_\epsilon^2]_j < +\infty$, for all $j \in [q]$.

Thus, by the above and Proposition 5, the class $\mathfrak{G}_\epsilon$ satisfies the assumptions in Theorem 5. Hence, $\mathfrak{G}_\epsilon$ is asymptotically continuous at $\dot{g}_{\theta_0}$.

Proof. We start by looking at the squared norm of the gradient at θ_0 , by (K.19)

$$\begin{aligned} F_u \|\dot{g}_{\theta_0}\|_2^2 &= F_u d^{-2} \|\mathbf{Z}_{t_i}^T \mathbf{u}_{t_i}\|_2^2 \\ &= d^{-2} F_u \{\mathbf{u}_{t_i}^T (\mathbf{Z}_t \mathbf{Z}_t^T) \mathbf{u}_{t_i}\} \\ &= d^{-2} F_u [\text{tr}\{\mathbf{u}_{t_i}^T (\mathbf{Z}_t \mathbf{Z}_t^T) \mathbf{u}_{t_i}\}] \\ &= d^{-2} \text{tr}[F_u \{\mathbf{u}_{t_i}^T (\mathbf{Z}_t \mathbf{Z}_t^T) \mathbf{u}_{t_i}\}] \\ &= d^{-2} \text{tr}(\mathbf{Z}_t \Sigma_u \mathbf{Z}_t^T) \\ &\leq d^{-1} \sigma_u^2 \text{tr}(\mathbf{Z}_t^T \mathbf{Z}_t) \end{aligned}$$

- By Proposition 2, there exists a $0 < \tau_{(q)} < +\infty$, which upper bounds all eigenvalues of $E(\mathbf{Z}_t^T \mathbf{Z}_t)$, so we have that

$$F_{\mathbf{y}_t, \mathbf{Z}_t} \|\dot{g}_{\theta_0}\|_2^2 \leq \sigma_u^2 \tau_{(q)} < +\infty. \quad (\text{K.26})$$

Let $\dot{g}_\theta \in \mathfrak{G}_\epsilon$, and notice that

$$\begin{aligned} \|\dot{g}_\theta\|_2^2 &= \|\dot{g}_\theta - \dot{g}_{\theta_0} + \dot{g}_{\theta_0}\|_2^2 \\ &\leq (\|\dot{g}_\theta - \dot{g}_{\theta_0}\|_2 + \|\dot{g}_{\theta_0}\|_2)^2 \\ &\leq 2 \left(\|\dot{g}_\theta - \dot{g}_{\theta_0}\|_2^2 + \|\dot{g}_{\theta_0}\|_2^2 \right). \end{aligned}$$

- By (K.22) and the above we have that

$$\|\dot{g}_\theta\|_2^2 \leq 2 \left(\|(\mathbf{Z}_t^T \mathbf{Z}_t)(\theta - \theta_0) + o(\|\theta - \theta_0\|_2)\|_2^2 + \|\dot{g}_{\theta_0}\|_2^2 \right).$$

- By taking expectations, (K.22) and (K.26), we arrive at

$$F_{\mathbf{y}_t, \mathbf{Z}_t} \|\dot{g}_\theta\|_2^2 \leq 4 \{(\epsilon^2 + \sigma_u^2) \tau_{(q)} + \epsilon^2\} < +\infty. \quad (\text{K.27})$$

By (K.27) we see that $F_{\mathbf{y}_t, \mathbf{Z}_t} \|G_\epsilon\|_2^2 < +\infty$, and noting that the ℓ_2 -norm of a vector is bounded if and only if all of its entries are bounded, we see that

$$F_{\mathbf{y}_t, \mathbf{Z}_t} [G_\epsilon]_j^2 < +\infty,$$

for all entries in G_ϵ .

We proceed to the second part. Notice that for all $j \in [q]$ each j th entry satisfies

$$\begin{aligned} [\dot{g}_\theta]_j &= -[\mathbf{Z}_{t_i}^T(\mathbf{y}_{t_i} - \mathbf{Z}_{t_i}\boldsymbol{\theta})]_j \\ &= -[\mathbf{Z}_{t_i}^T\mathbf{y}_{t_i} - (\mathbf{Z}_{t_i}^T\mathbf{Z}_{t_i})\boldsymbol{\theta}]_j \end{aligned}$$

Let $\phi_{0,j} \in \mathbb{R}^{q+1}$ such that $\phi_{0,j} = -[\mathbf{Z}_{t_i}^T\mathbf{y}_{t_i}]_j$ and all other entries are $\phi_j = [(\mathbf{Z}_{t_i}^T\mathbf{Z}_{t_i})]_{j\cdot}$, i.e., the j th column in $(\mathbf{Z}_{t_i}^T\mathbf{Z}_{t_i})$. Then,

$$[\dot{g}_\theta]_j = \phi_{0,j} + \langle \phi_j, \boldsymbol{\theta} \rangle,$$

for all $j \in [q]$, i.e., each entry is a hyperplane in $\mathbb{R}^{q+1}$. Thus, it is a finite dimension Vapnik-Chervonenkis graph class because the subgraph corresponds to the subgraph of a hyperplane.

Recall that an intersection of Vapnik-Chervonenkis graph classes is a Vapnik-Chervonenkis graph class. Thus, all j th entries in $\dot{g}_\theta$ are finite dimension Vapnik-Chervonenkis graph classes. Hence, $\mathfrak{G}_\epsilon$ is a finite dimension Vapnik-Chervonenkis graph class; see van de Geer (2020).

Therefore, by combining the above, (K.27) and (K.25), we see that $\mathfrak{G}_\epsilon$ satisfies the assumptions in Theorem 5. Hence, $\mathfrak{G}_\epsilon$ is asymptotically continuous at $\dot{g}_{\theta_0}$. $\blacksquare$

K.3.2 Completion of asymptotic normality

We are ready to finish our proof of asymptotic normality.

Proof. By Propositions 2–6 we see that $\hat{\boldsymbol{\theta}}$ satisfies the assumptions in Lemma 6. Define the following $\boldsymbol{\Gamma}(p, [C]) := E\{(\mathbf{Z}_t^T\mathbf{Z}_t)\}$. Then, by (K.22) we see that the influence function is $d^{-1}\{\boldsymbol{\Gamma}(p, [C])\}^{-1}\dot{g}_{\theta_0}$. Notice that

$$\begin{aligned} F_{\mathbf{y}_t, \mathbf{Z}_t}(\dot{g}_{\theta_0}\dot{g}_{\theta_0}^T) &= F_{\mathbf{Z}_t}\{F_{\mathbf{u}}(\dot{g}_{\theta_0}\dot{g}_{\theta_0}^T)\} \\ &= F_{\mathbf{Z}_t}d^{-2}\{F_{\mathbf{u}}(\mathbf{Z}_{t_i}^T\mathbf{u}_{t_i}\mathbf{u}_{t_i}^T\mathbf{Z}_{t_i})\} \\ &= F_{\mathbf{Z}_t}d^{-2}\{\mathbf{Z}_{t_i}^T\boldsymbol{\Sigma}_{\mathbf{u}}\mathbf{Z}_{t_i}\}, \end{aligned}$$

by Lemma 6 we see that the asymptotic covariance is

$$\boldsymbol{\Sigma}_{\hat{\boldsymbol{\theta}}} = d^{-1}\{\boldsymbol{\Gamma}(p, [C])\}^{-1}E\{(\mathbf{Z}_t^T\boldsymbol{\Sigma}_{\mathbf{u}}\mathbf{Z}_t)\}\{\boldsymbol{\Gamma}(p, [C])\}^{-1},$$

where, we send the other d^{-1} term to the sample size divisor. Finally, by recalling that $|\mathcal{K}| = d$ and applying Lemma 6 we have that

$$\{|\mathcal{K}|(T - p)\}^{1/2}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N_q(0, \boldsymbol{\Sigma}_{\hat{\boldsymbol{\theta}}}).$$

$\blacksquare$

Proposition 7. *Assume that A.1, A.2, and A.3 hold in Assumptions 1. Further, assume that all conditions in Proposition 3 hold, and that $\mathbf{u}_t \sim \text{WN}(0, \sigma_{\mathbf{u}}^2\mathbf{I}_d)$, i.e., the white noise processes are i.i.d. at all times t . Then, we have that*

$$\hat{\sigma}_{\mathbf{u}}^2 = \{|\mathcal{K}|(T - p)\}^{-1}\|\mathbf{y} - \mathbf{R}\hat{\boldsymbol{\theta}}\|_2^2,$$

is a consistent estimator of $\sigma_{\mathbf{u}}^2$, i.e., $\hat{\sigma}_{\mathbf{u}}^2 \xrightarrow{\text{Pr}} \sigma_{\mathbf{u}}^2$.

Proof. By Proposition 3 and the continuous mapping theorem, as $\hat{\boldsymbol{\theta}} \xrightarrow{\text{Pr}} \boldsymbol{\theta}_0$ we have that

$$\begin{aligned} \{|\mathcal{K}|(T-p)\}^{-1} \|\mathbf{y} - \mathbf{R}\hat{\boldsymbol{\theta}}\|_2^2 &\xrightarrow{\text{Pr}} \{|\mathcal{K}|(T-p)\}^{-1} \|\mathbf{y} - \mathbf{R}\boldsymbol{\theta}_0\|_2^2 \\ &= \{|\mathcal{K}|(T-p)\}^{-1} \|\mathbf{u}\|_2^2. \end{aligned}$$

Notice that for all entries $u_{i,t}$ in $\mathbf{u}$, we have that $E(u_{i,t}^2) = \sigma_{\mathbf{u}}^2$. Since, the above is the sample mean of $\{|\mathcal{K}|(T-p)\}$ *i.i.d.* realisations of $u_{i,t}^2$, by the (weak) law of large numbers, we have that

$$\{|\mathcal{K}|(T-p)\}^{-1} \|\mathbf{u}\|_2^2 \xrightarrow{\text{Pr}} \sigma_{\mathbf{u}}^2,$$

whence, we arrive at

$$\hat{\sigma}_{\mathbf{u}}^2 \xrightarrow{\text{Pr}} \sigma_{\mathbf{u}}^2.$$

■

L Proof of results in Section 2.1